



At the Intersection of Biology, Statistics and Computer Science

Courtney Montgomery, PhD

Director, Center for Biomedical Data Sciences

Professor, Genes and Human Disease

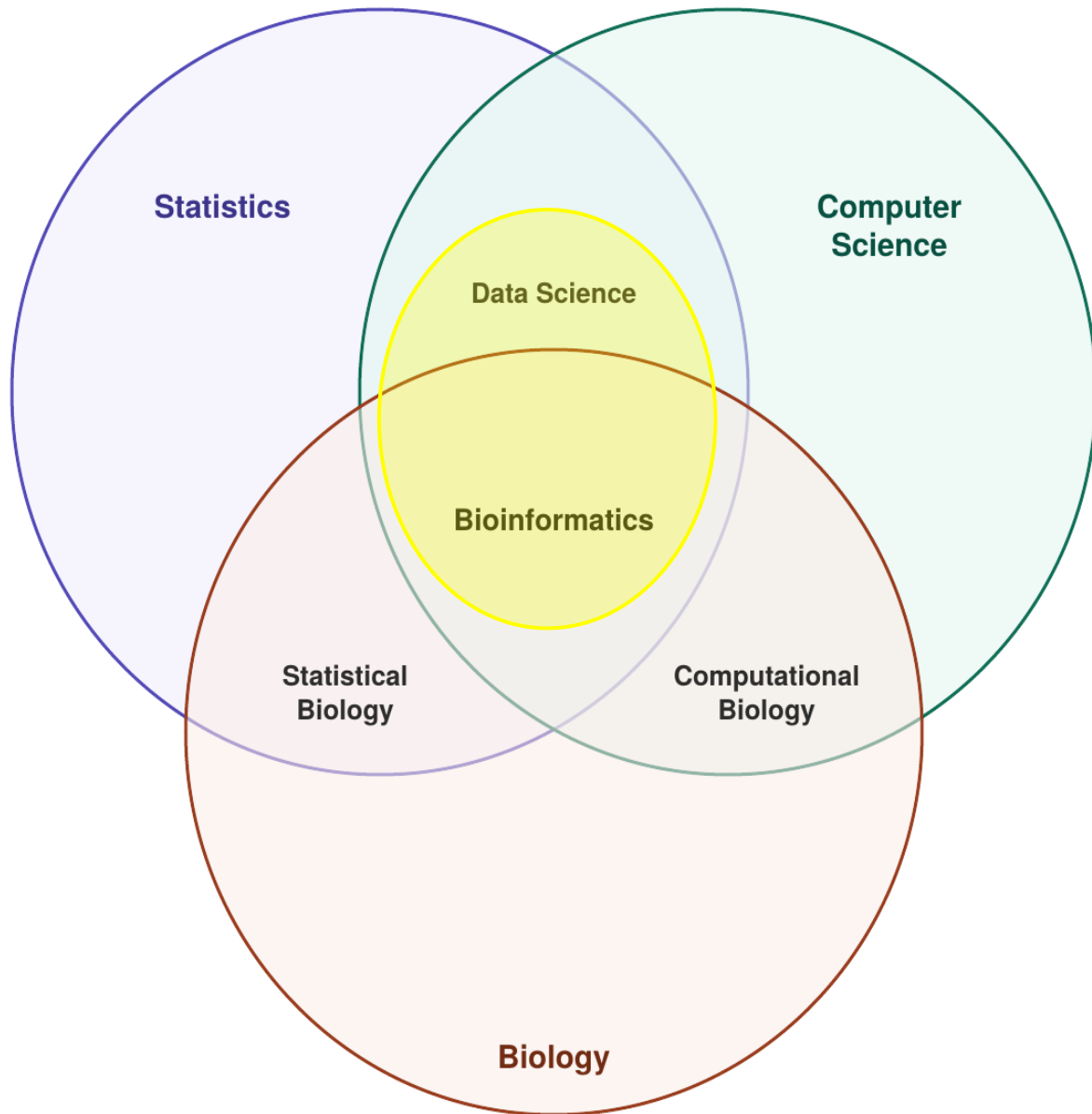
Department of Microbiology and Immunology, OUHealth

Department of Pathology, OUHealth

Department of Biostatistics and Epidemiology, OUHealth

School of Computer Science, OU

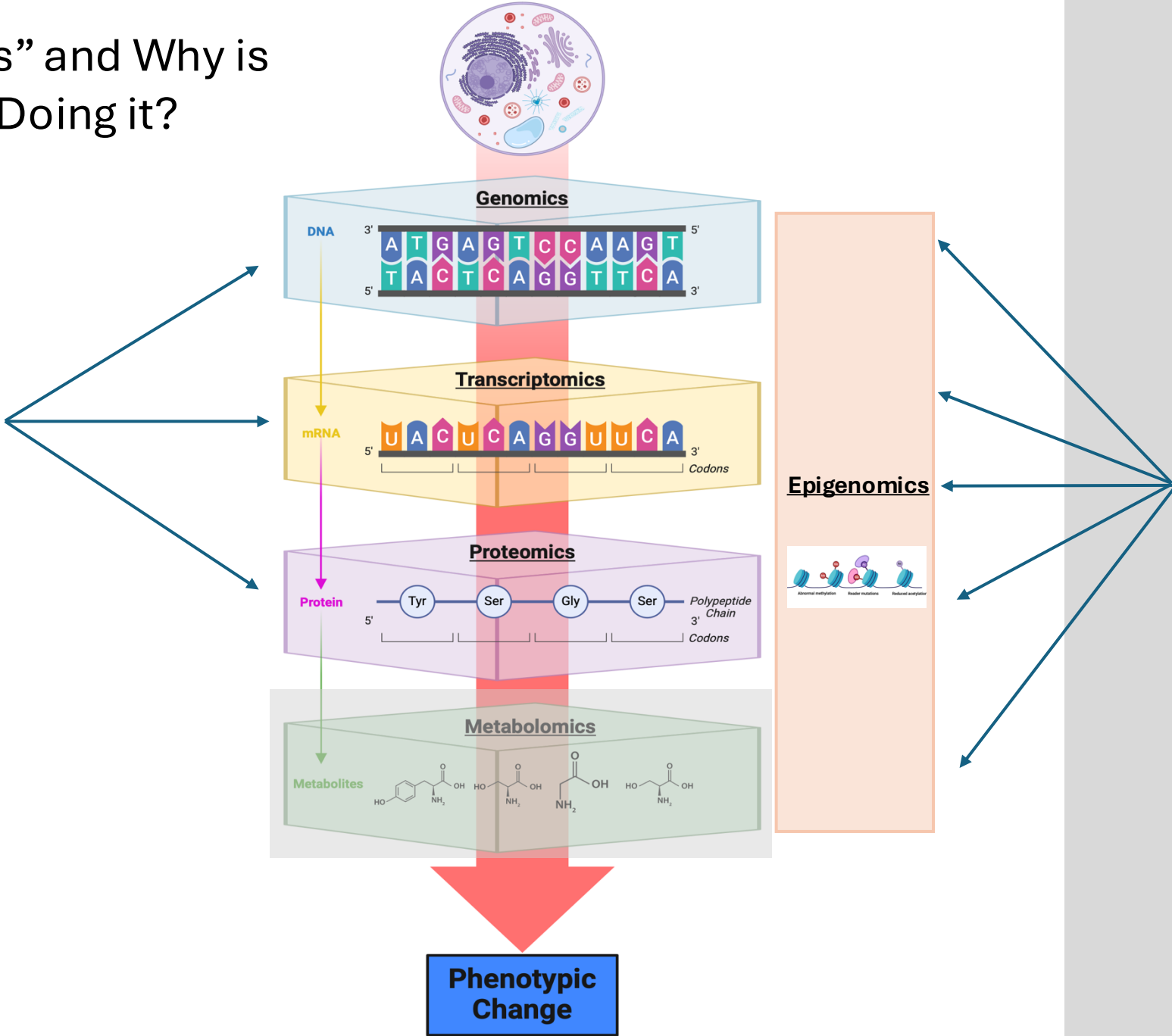




- Technological Advancements have led to significant advances in multiple areas:
 - “Omics” / NGS sequencing
 - Microscopy
 - Computational speed

What is “omics” and Why is Everyone Doing it?

The Central Dogma
THEN



The Central Dogma
NOW

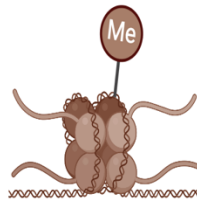
NGS Technologies

Greatly facilitates all Omics research, some better than others

Next Generation Sequencing Applications

Epigenetics

DNA methylation analysis



- Tissue identification
- Twins differentiation
- Prediction of age
- Drug consumption

MicroRNA

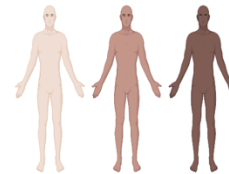
RNA analysis



- Tissue identification
- Post-mortem interval
- Prediction of stain deposition time

Phenotype

SNPs analysis



- Skin, hair, iris color
- Baldness
- Age

Ancestry

SNPs analysis



- Ethnic difference
- Biogeographic origin
- Population analysis

Non-human identification

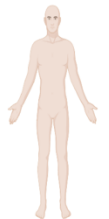
STRs/SNPs analysis



- Microbiome
- Animal/plant species analysis

Human identification

STRs/SNPs/MLIs analysis



- Identity
- Kinship
- Mix analysis

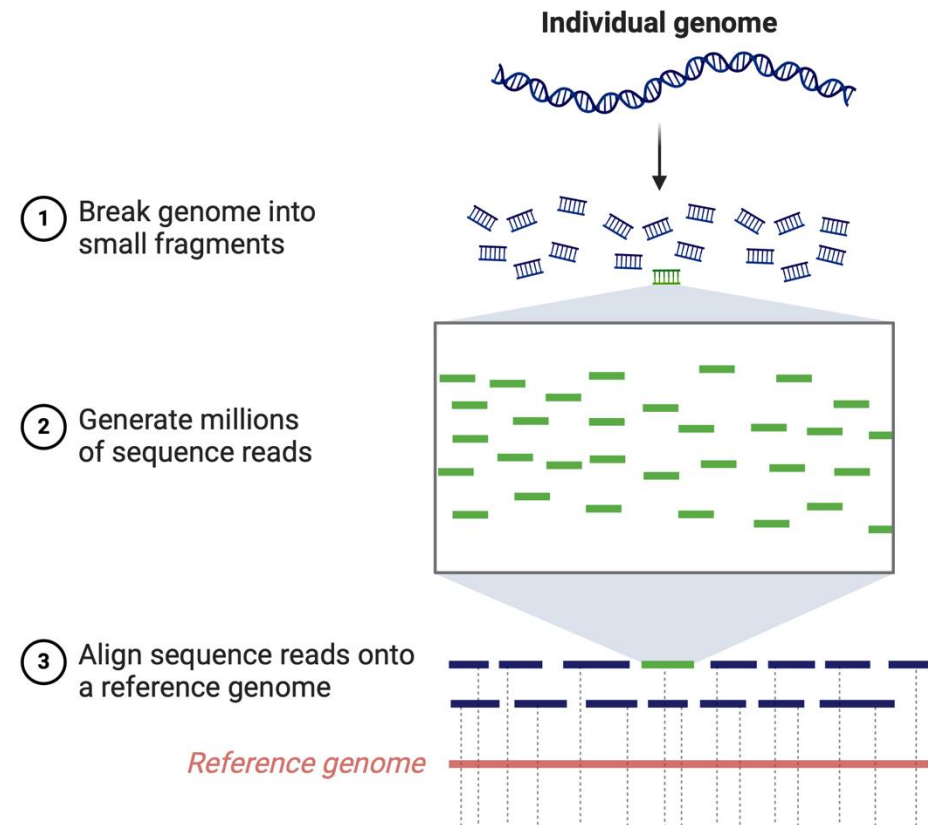
Predictiveness

Accuracy

Genetics (Genomics)

- Genotyping
 - Linkage
 - Association
- DNA sequencing
 - Whole exome
 - Targeted
 - Whole genome

Sequencing an Individual Genome



Genomics: Association Analysis

CAGTATCGTAATATAATAATAAGTCCGGAT
GTCATAGCATTATATTATTATTCAGGCTA

CAGTATCGTAATATAATAATAAGTCCGGAT
GTCATAGCATTATATTATTATTCAGGCTA

CAGTATCGTAATATAATAATAAGTCCGGAT
GTCATAGCATTATATTATTATTCAGGCTA

CAGTATCGTAATATAATAATAAGTCCGGAT
GTCATAGCATTATATTATTATTCAGGCTA

With Disease

CAGTATCGTAATATAATAATAAGTCCGGAT
GTCATAGCATTATATTATTATTCAGGCTA

CAGTATCGTAATATAATAATAAGTCCGGAT
GTCATAGCATTATATTATTATTCAGGCTA

CAGTATCGTAATATAATAATAAGTCCGGAT
GTCATAGCATTATATTATTATTCAGGCTA

CAGTATCGTAATATAATAATAAGTCCGGAT
GTCATAGCATTATATTATTATTCAGGCTA

CAGTATCGTAATATAATAATAAGTCCGGAT
GTCATAGCATTATATTATTATTCAGGCTA

w/o Disease

CAGTATCGTAATATAATAATAAGTCCAGAT
GTCATAGCATTATATTATTATTCAGGCTA

CAGTATCGTAATATAATAATAAGTCCAGAT
GTCATAGCATTATATTATTATTCAGGCTA

CAGTATCGTAATATAATAATAAGTCCAGAT
GTCATAGCATTATATTATTATTCAGGCTA

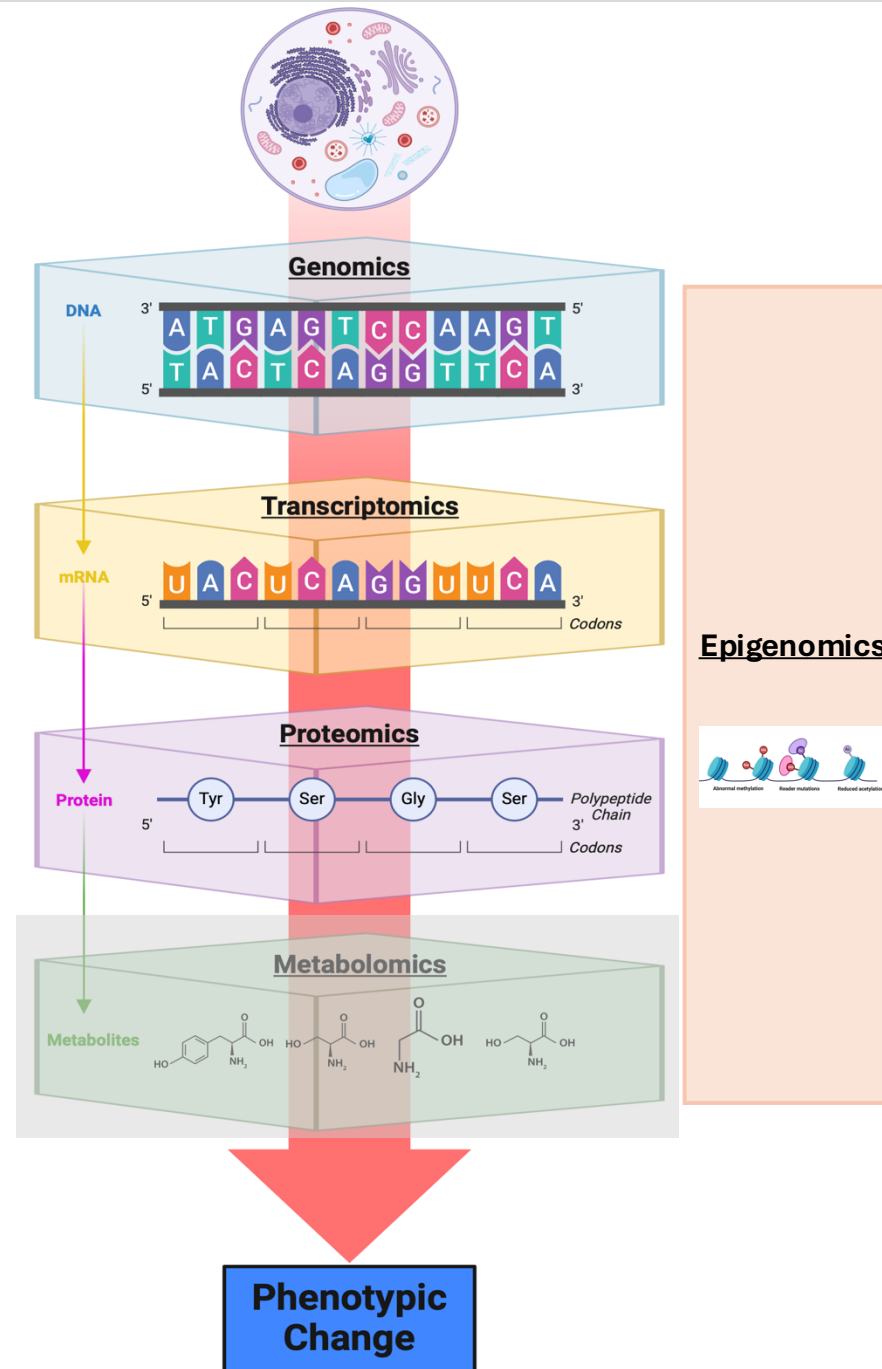
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GTCATAGCATTATATTATTATTCAGGCTA

CAGTATCGTAATATAATAATAAGTCCAGAT
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CAGTATCGTAATATAATAATAAGTCCAGAT
GTCATAGCATTATATTATTATTCAGGCTA

CAGTATCGTAATATAATAATAAGTCCAGAT
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CAGTATCGTAATATAATAATAAGTCCAGAT
GTCATAGCATTATATTATTATTCAGGCTA

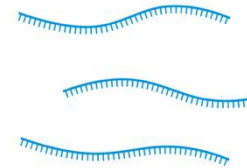


Transcriptomics

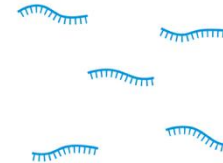
Bulk sequencing

RNA Sequencing

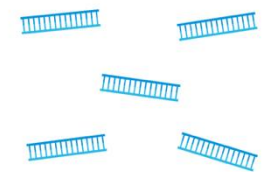
① Isolate RNA from samples



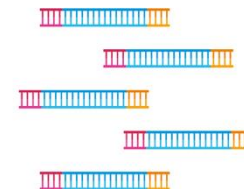
② Fragment RNA into short segments



③ Convert RNA fragments into cDNA



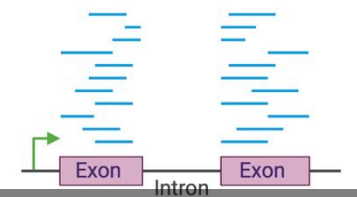
④ Ligate sequencing adapters and amplify



⑤ Perform NGS sequencing

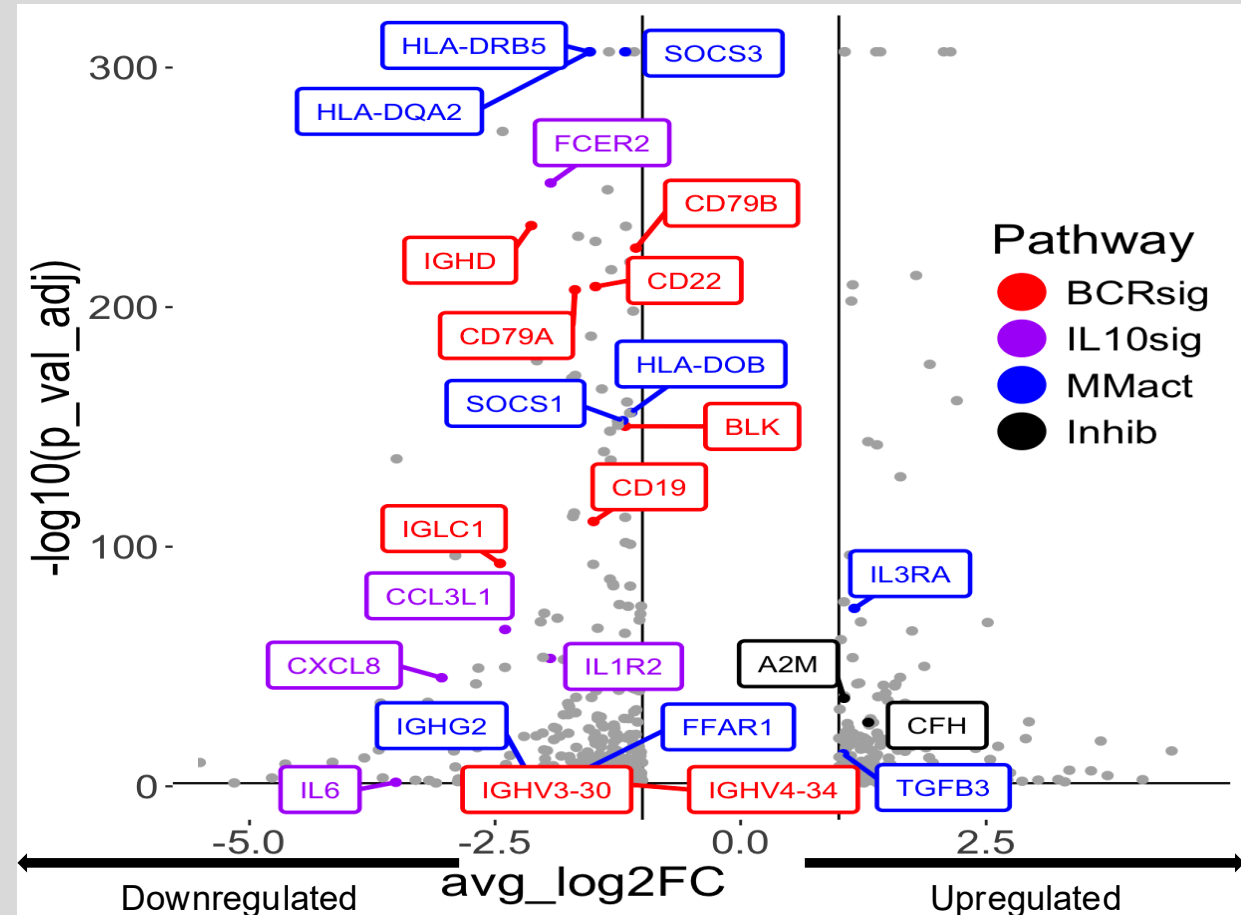


⑥ Map sequencing reads to the transcriptome/genome



Transcriptomics: Bulk Analysis

- **Transcriptomic characterization of HLA-DRB1*03:01 carriers.** Statistically significant differentially expressed genes when comparing carriers of HLA-DRB1*03:01 to non-carriers, genes ($\log_2FC > |1|$, FDR corrected p-value < 0.05). FDR corrected p-values of near zero were set to the smallest measurable adjusted p-value for visualization purposes. BCRsig = B cell receptor signaling pathway, IL10sig = IL-10 signaling pathway, MMact = Monocyte/Macrophage activation pathway, Inhib = inflammation inhibitory genes.



Transcriptomics: Bulk, Pros, Cons, and “Work Arounds”

PROS:

- Relatively easy prep
- Low minimum sample
- Results may be generalizable
- Affordable

CONS:

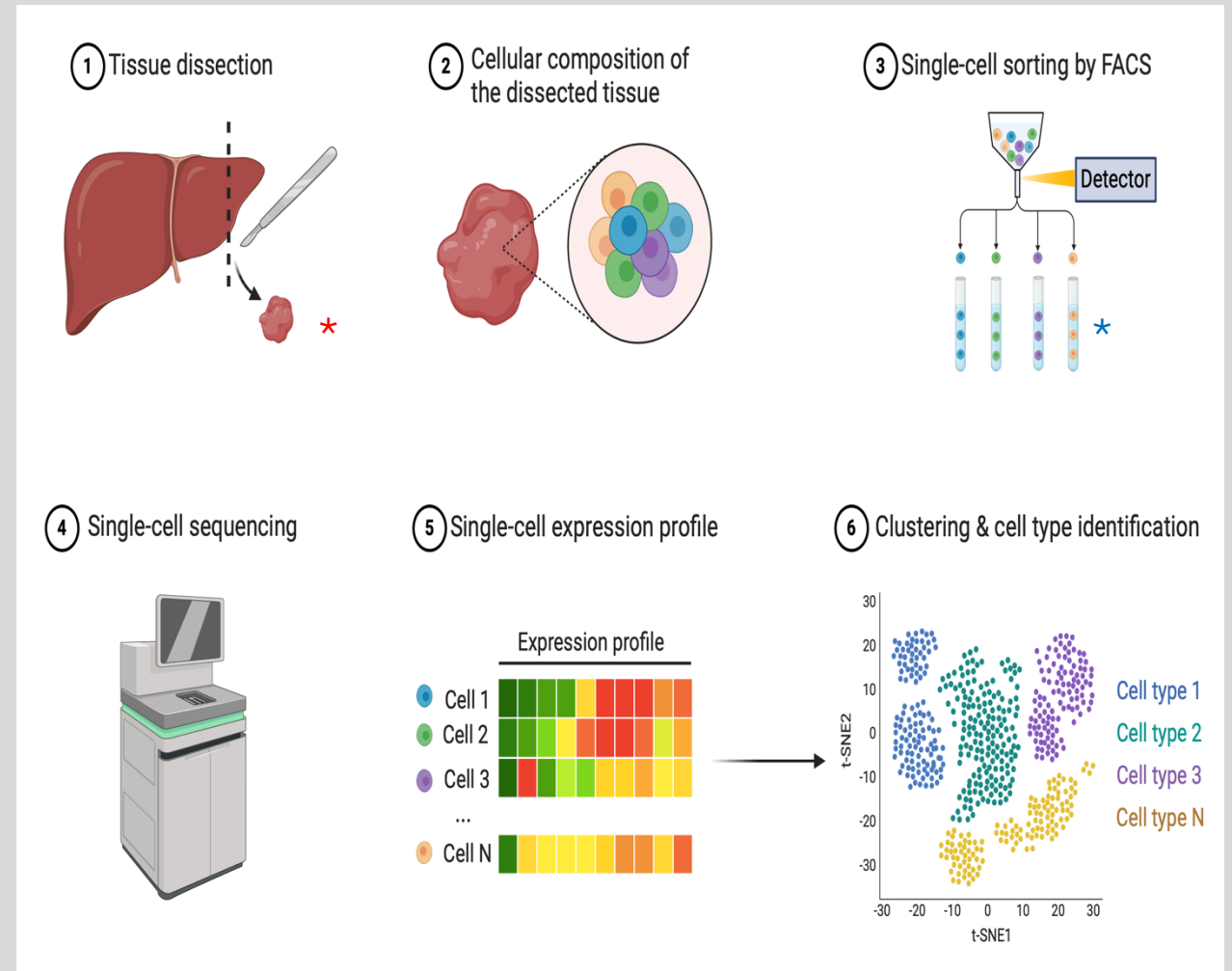
- Heterogeneity, depending on sample type
- Low Power - Unit of observation is the sample
- Hard to interpret

“Work-arounds”:

- Pre-sorting – may affect expression
- Deconvolution

Single Cell sequencing Technology

Single Cell Transcriptomics

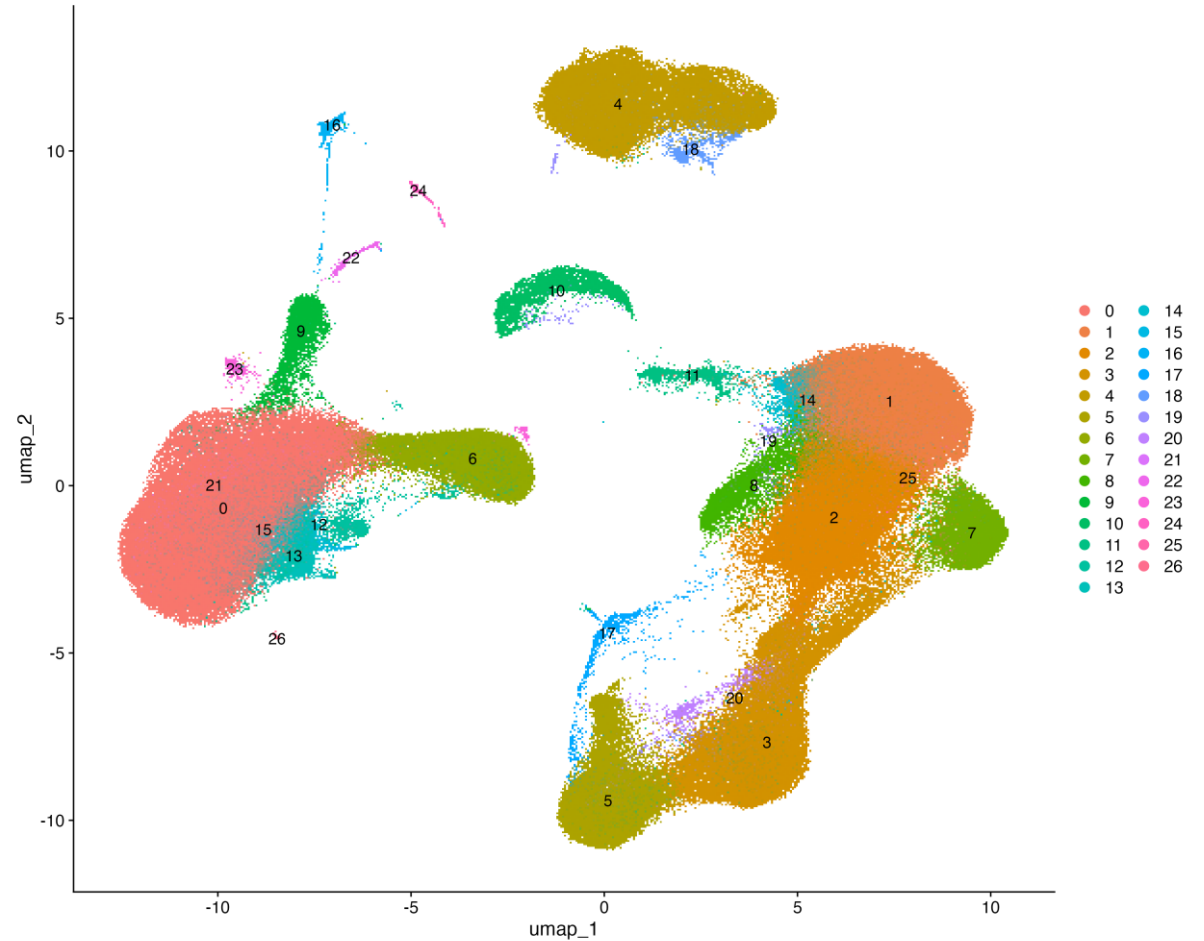


* Tissue includes whole blood, PBMCs, etc

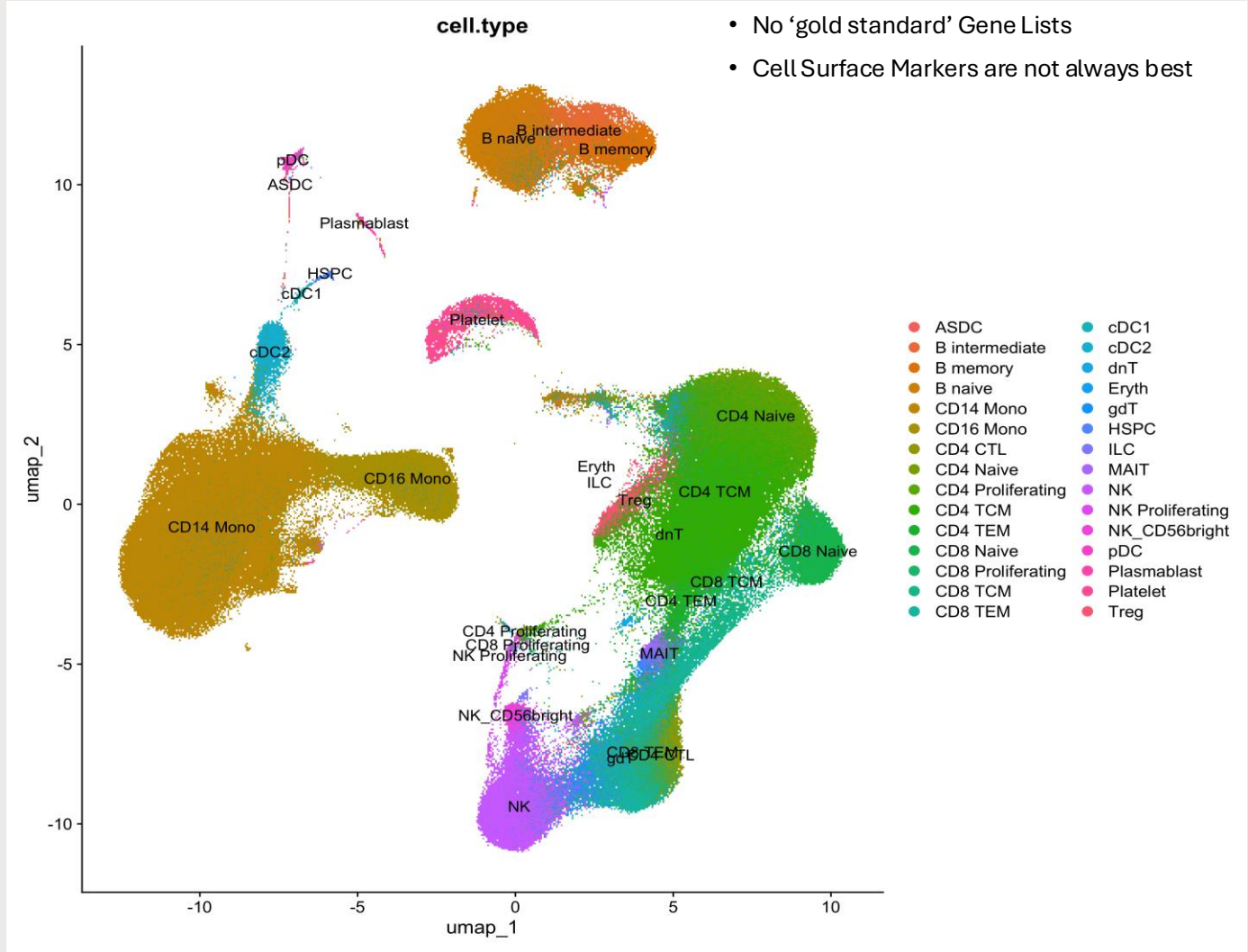
* 10x uses a gel bead emulsion prep (GEMs) for lysis and amplification of barcoded RNA

Single Cell Transcriptomics: Create Clusters

Uniform Manifold
Approximation and
Projection (UMAP)



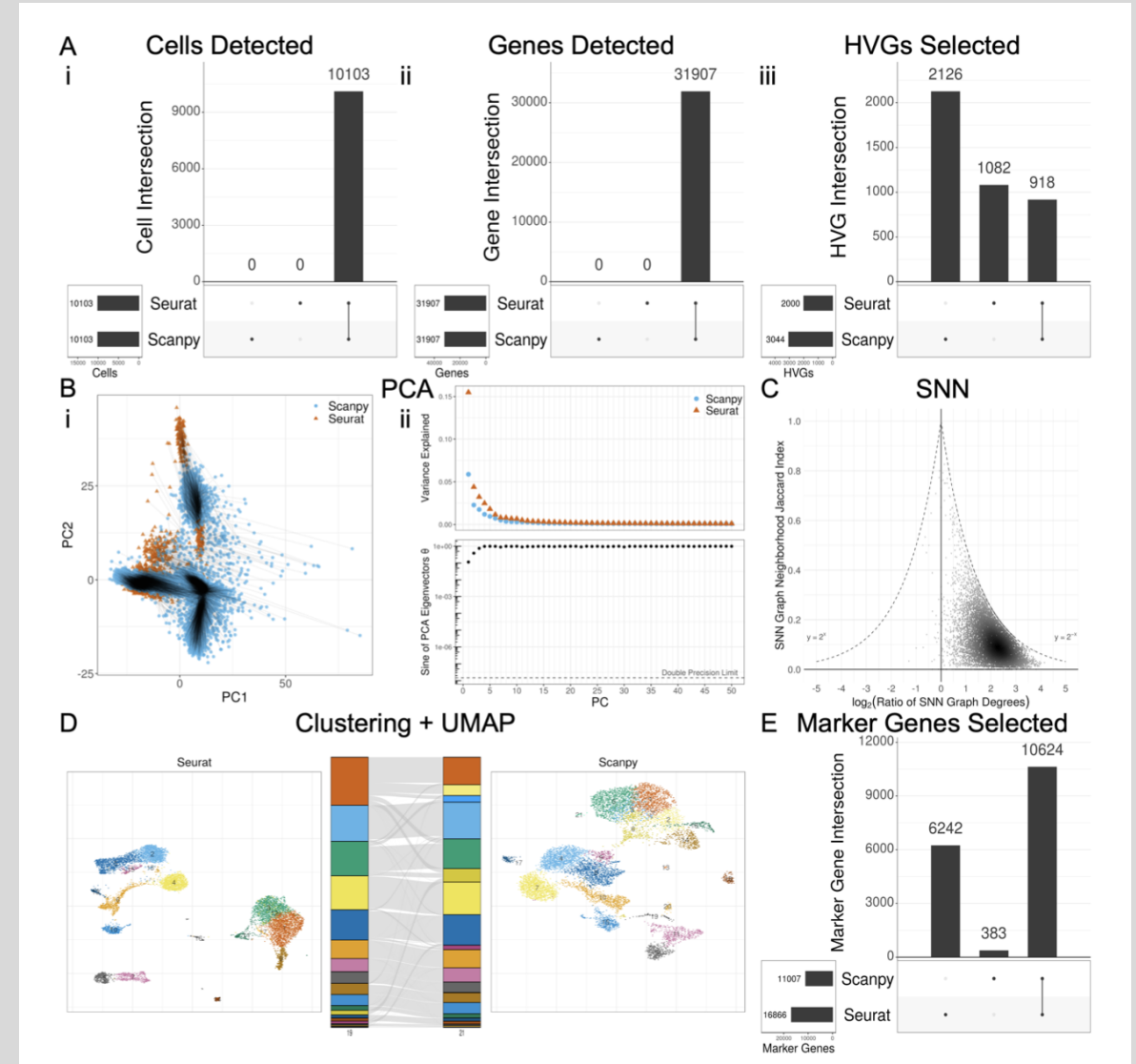
Single Cell Transcriptomics: Cell type Assignment



scGeneFit: Dumitrascu, B., D.G. et al. Optimal marker gene selection for cell type discrimination in single cell analyses. *Nat Commun* **12**, 1186 (2021).

Single Cell Analysis: An Art and a Science

- Seurat v. ScanPy
- Take-home Message: Seurat and ScanPy show considerable differences in scRNA-seq workflow results with default function arguments, BUT if parameters are chosen based on *a priori* knowledge and are **reasonable**, they are quite similar



Rich JM, et al. The impact of package selection and versioning on single-cell RNA-seq analysis. bioRxiv [Preprint]. 2024 Apr 11:2024.04.04.588111.

Transcriptomics: Single Cell, Pros, Cons, and “Work Arounds”

PROS:

- Cell-type specific results
- Sample is now the cell – but not independent
- Effects of prep significantly reduced
- Almost endless ways to subset the data

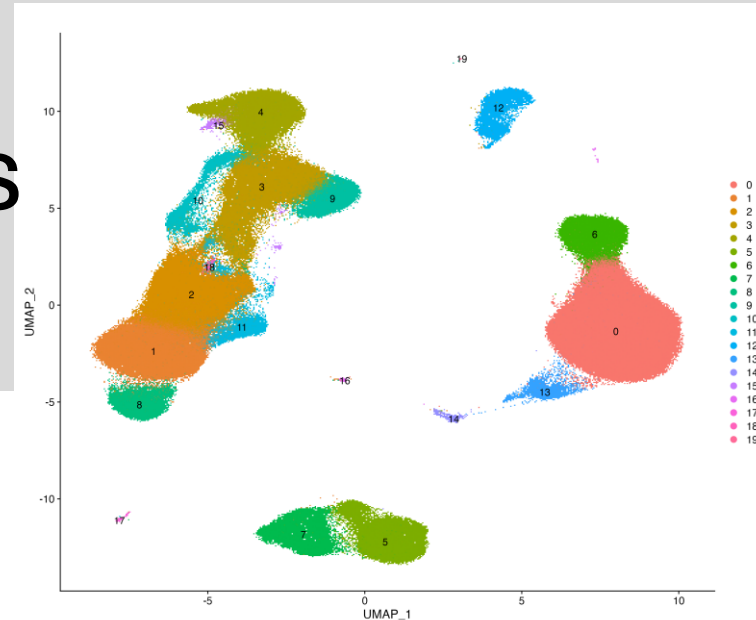
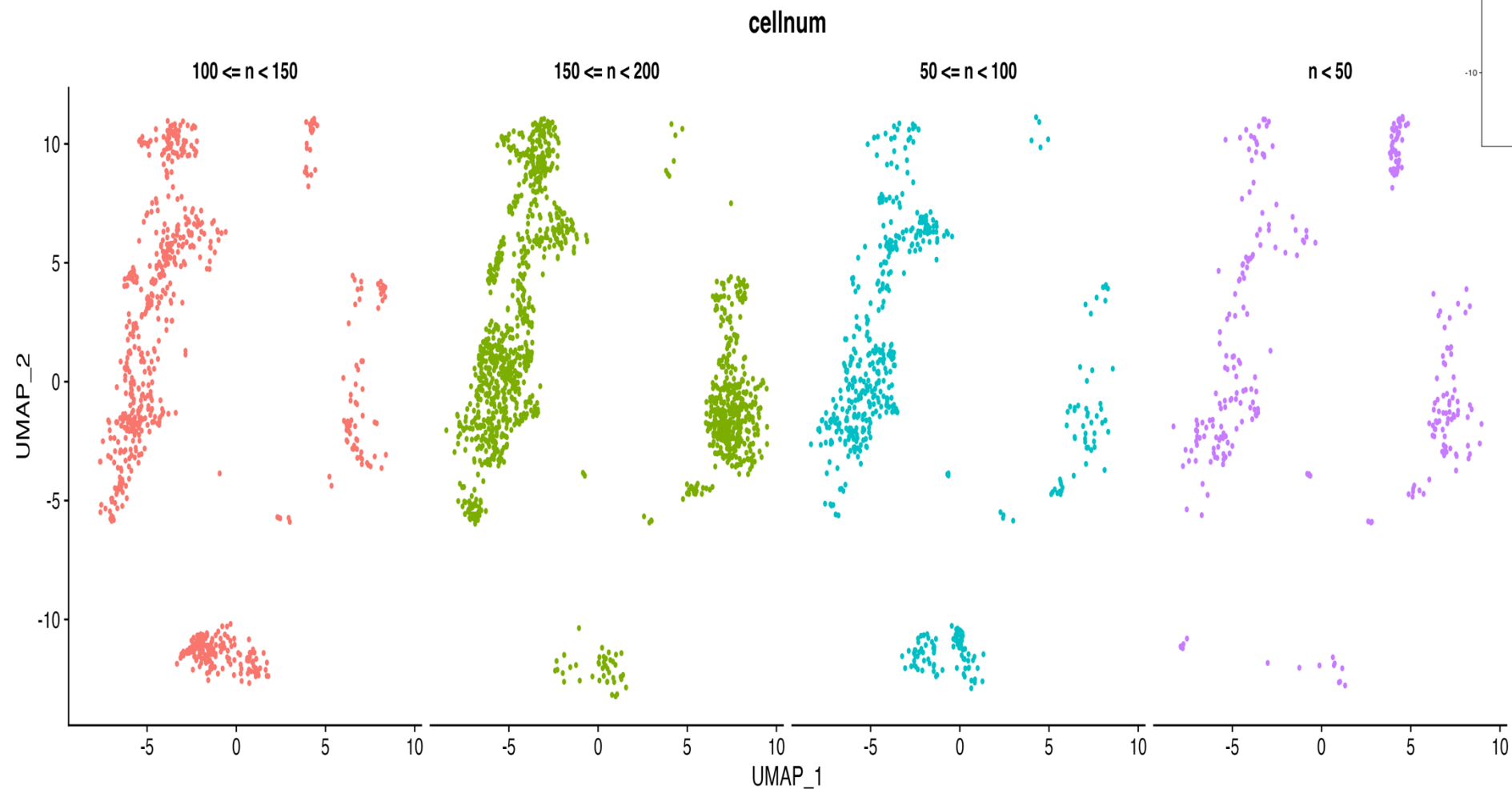
CONS:

- Expensive
- Still need reasonable sample sizes – especially if interested in rare cell types
- Batch effects
- Almost endless ways to subset the data

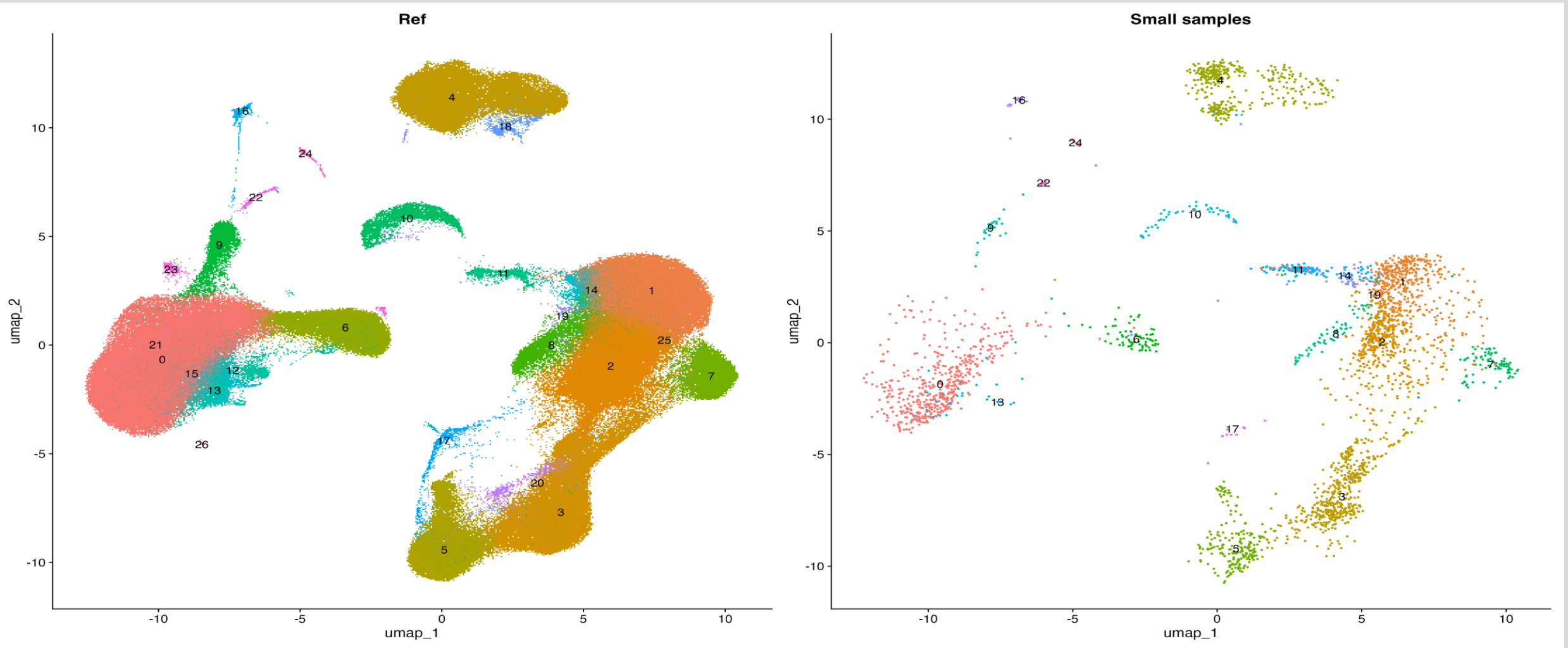
“Work-arounds”:

- Can still do pre-sorting – may affect expression
- Deconvolution
- Harmonization

Single Cell Transcriptomics: Batches

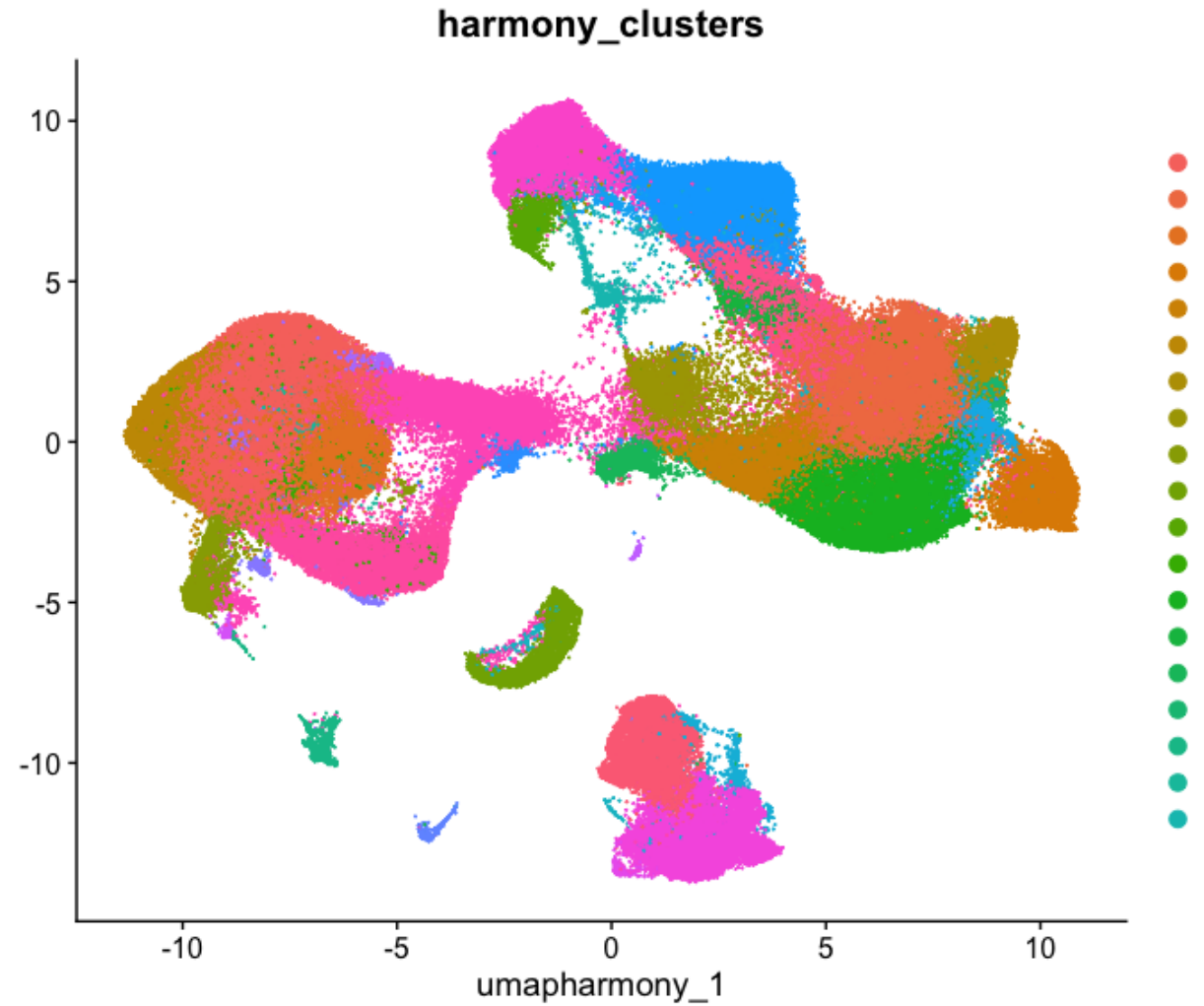


Single Cell Transcriptomics: Batches

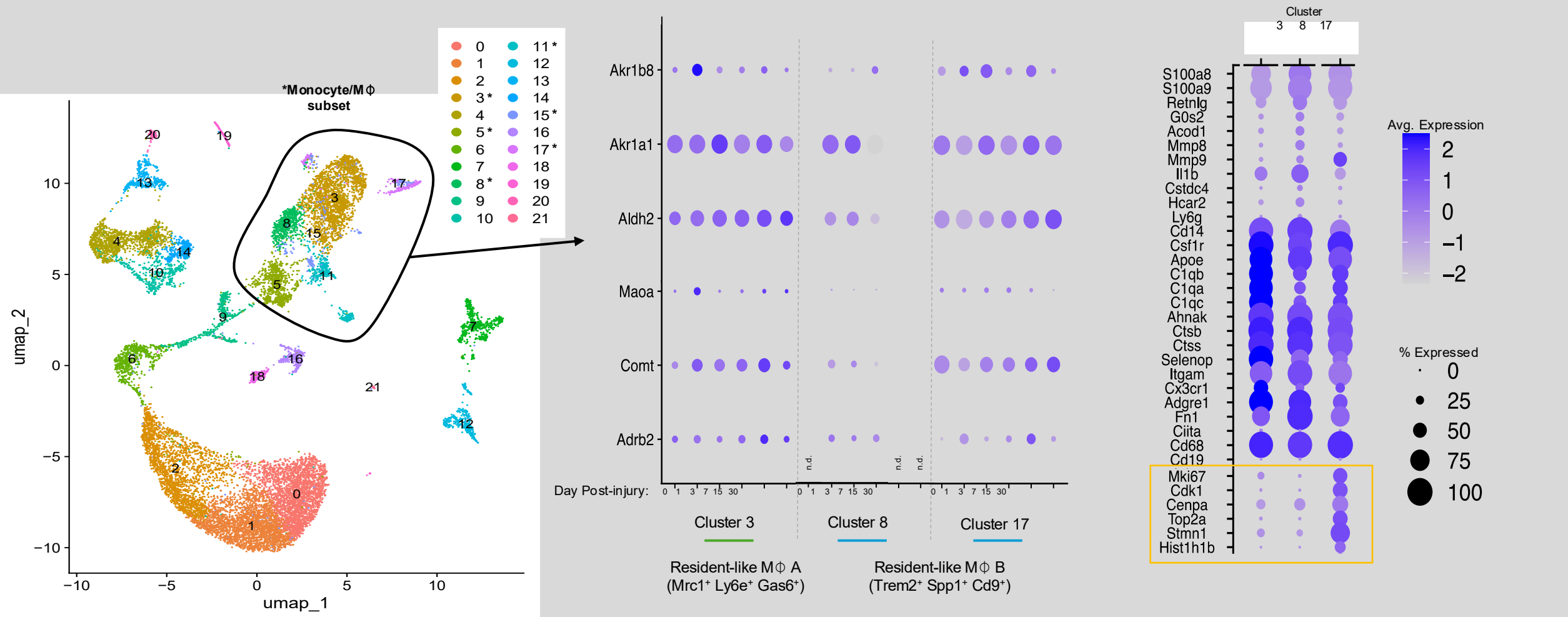


Single Cell Transcriptomics: Harmonization

Harmonize across any
number of parameters



Single-Cell Transcriptomics: Using Public Data to Identify Cellular Subpopulations of Interest



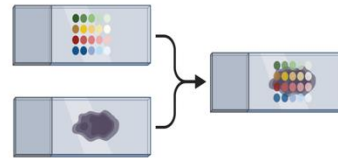
Komaravolu RK, et al. Sex-specific effects of injury and beta-adrenergic activation on metabolic and inflammatory mediators in a murine model of post-traumatic osteoarthritis. Osteoarthritis Cartilage. 2024 Mar 26:S1063-4584(24)01125-7.

Spatial Transcriptomics

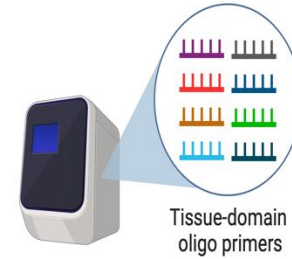
Spatial Transcriptomics

1 Tissue analysis

Grid tissue section on oligo(dT) primers slide

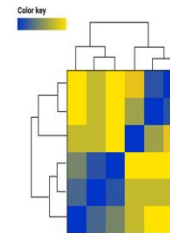


2 cDNA synthesis



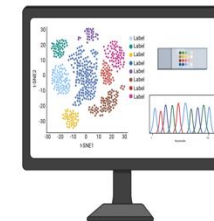
3 Gene expression

Spatially-resolved genes encoded via oligo tag



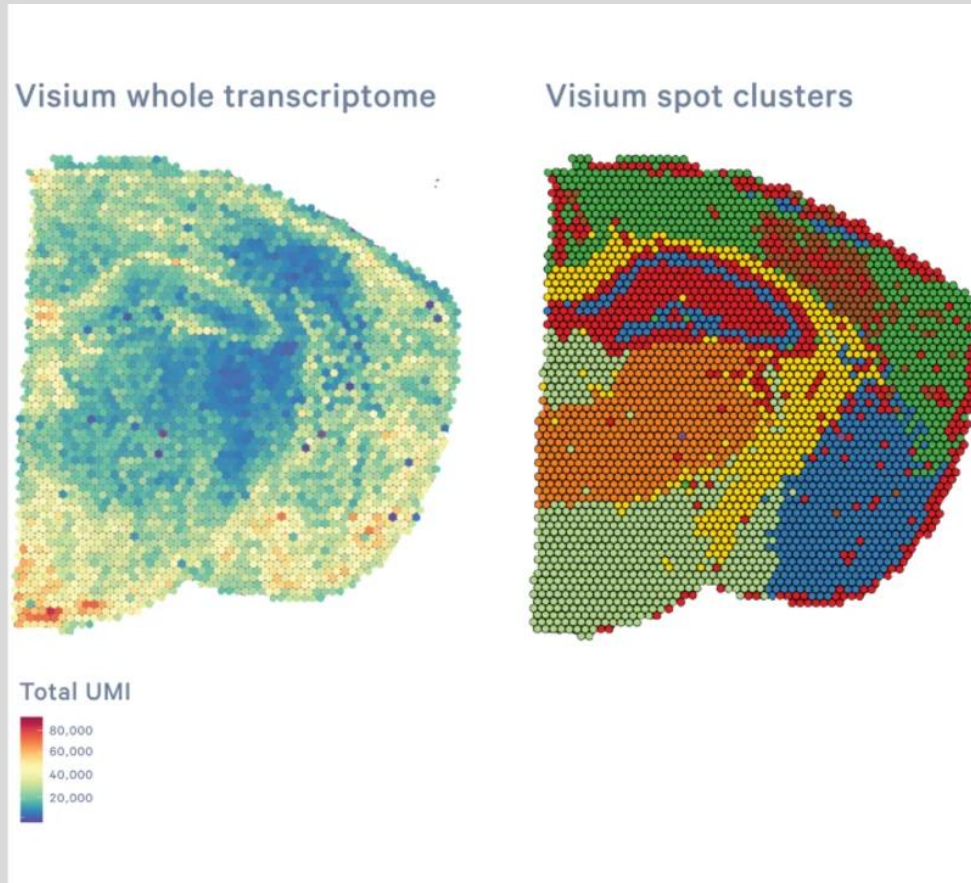
4 Data analysis

Gene clusters association to tissue domains

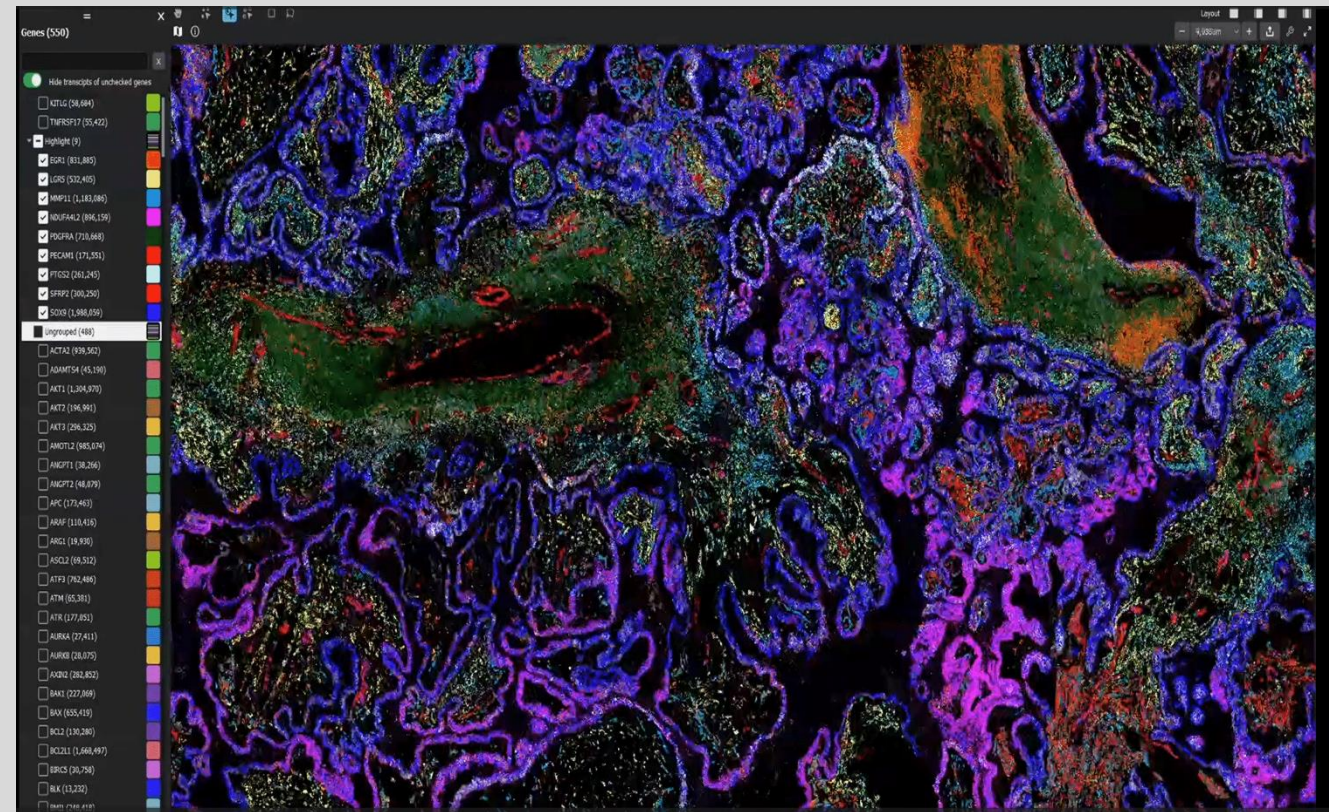


Spatial Transcriptomics

- Visium (10x) v. Xenium (10x) and MERSCOPE/Merfish (Vizgen)



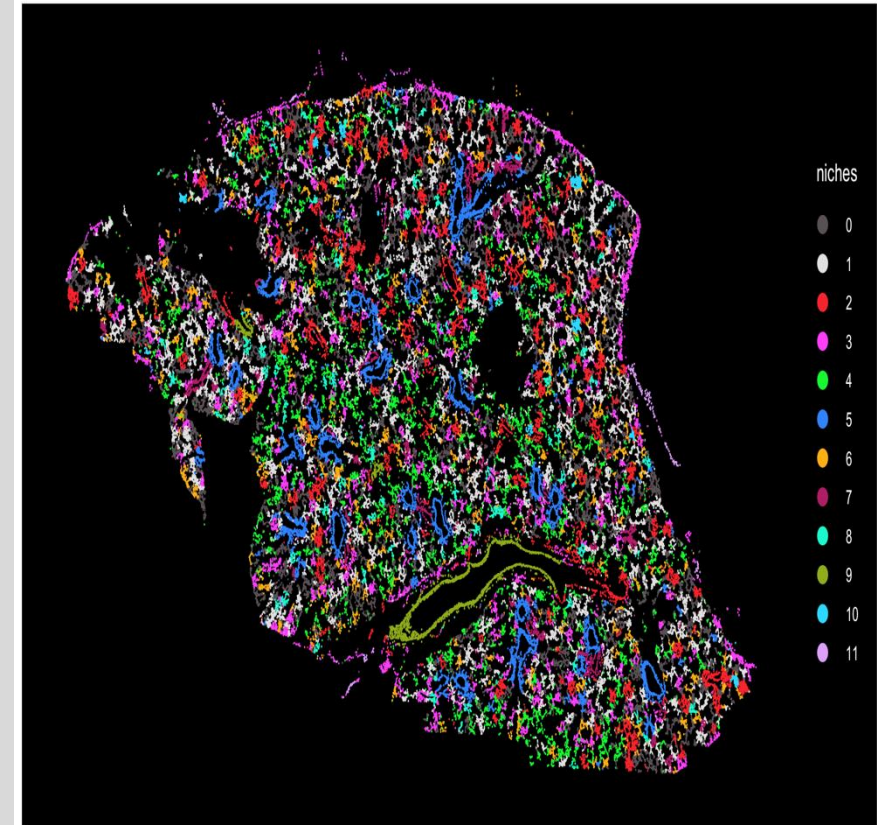
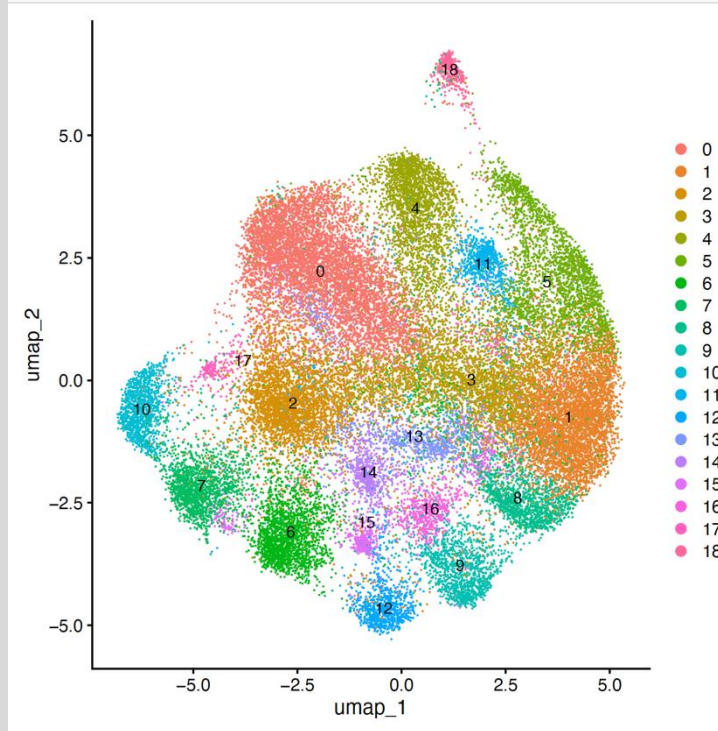
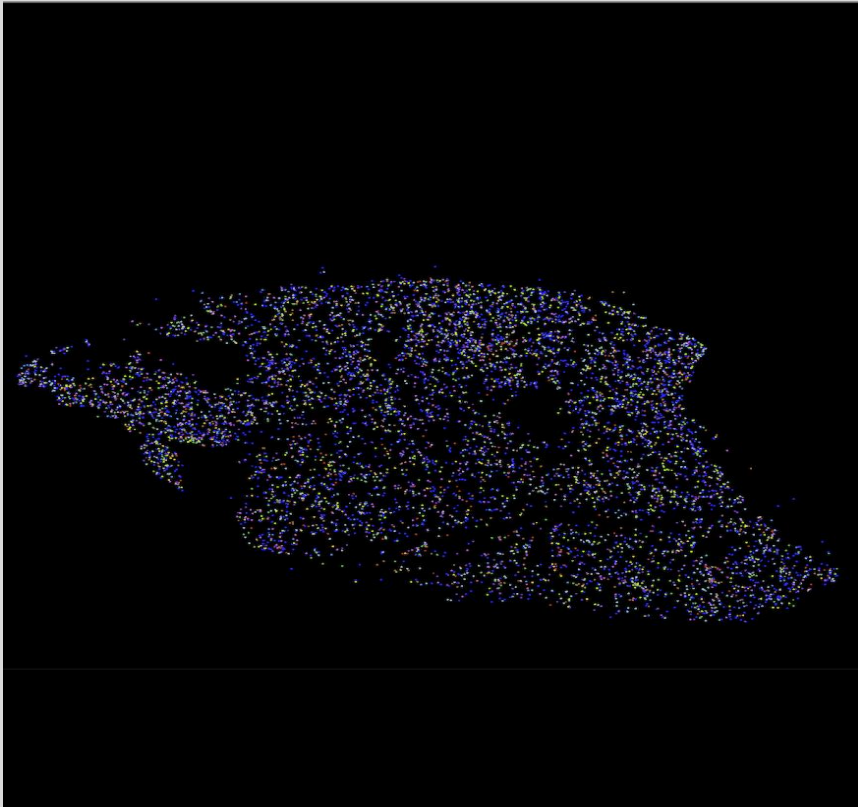
Via – Vizgen MERSCOPE Platform



Visium Spatial Platform - 10x Genomics

Spatial Transcriptomics: Clustering

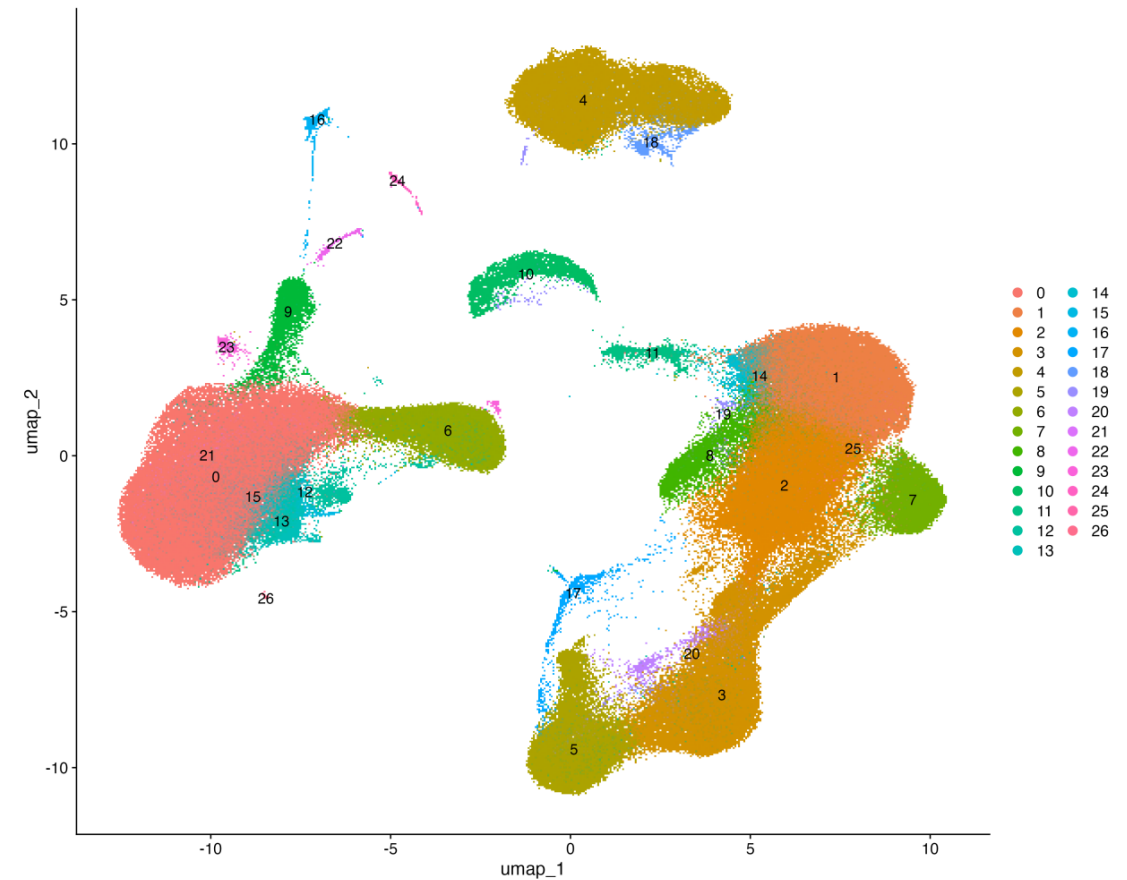
QC, Cluster, and Evaluate Cellular Neighborhoods in Single Cell Spatial Transcriptomic Data



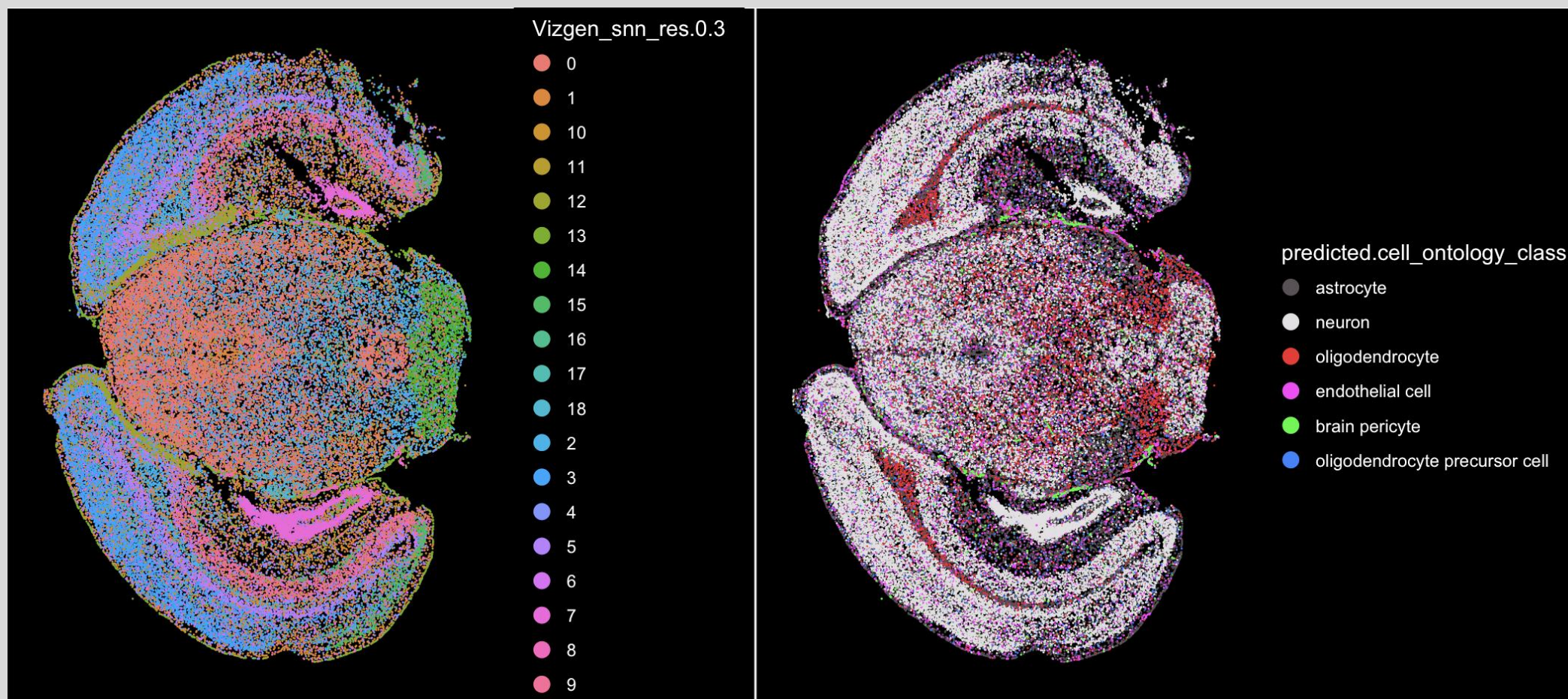
Data Courtesy of
Pengchun Yu, Irma Gryniuk

Single Cell / Spatial Integration

Have location need cell type

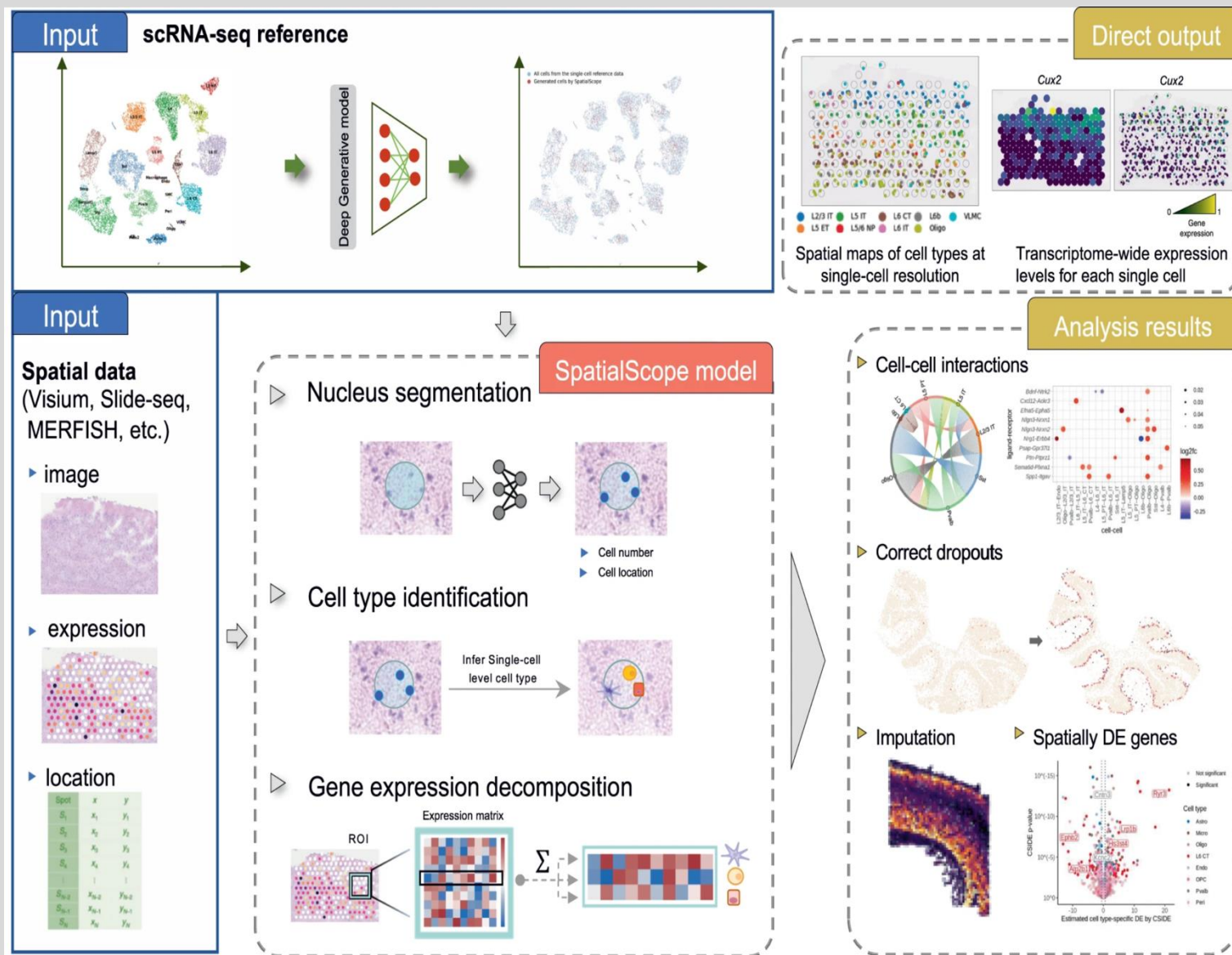


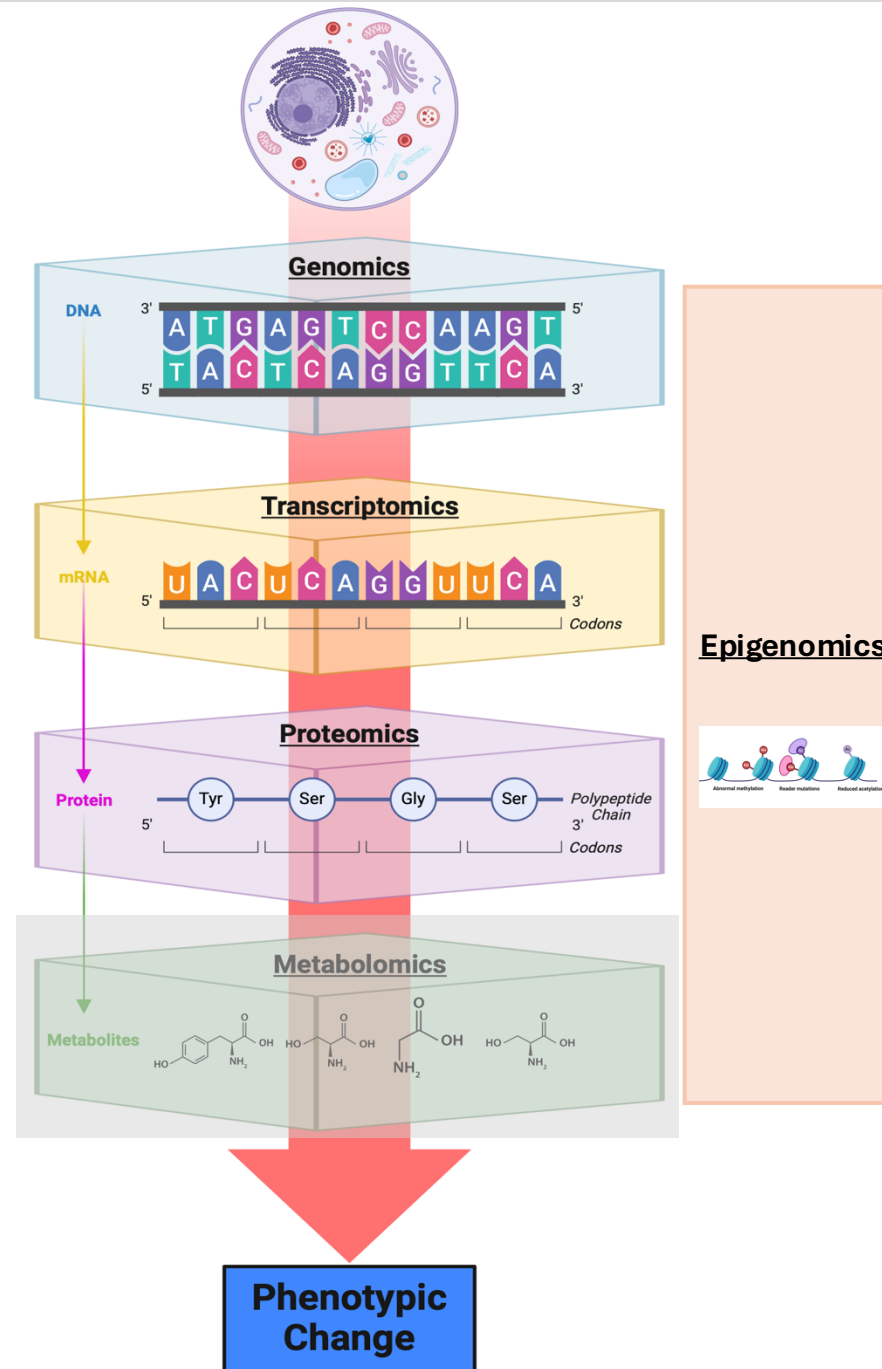
Single Cell / Spatial Integration

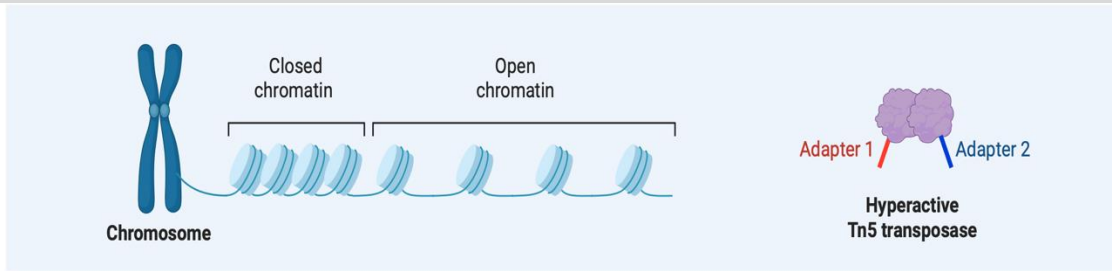


Single Cell / Spatial Integration

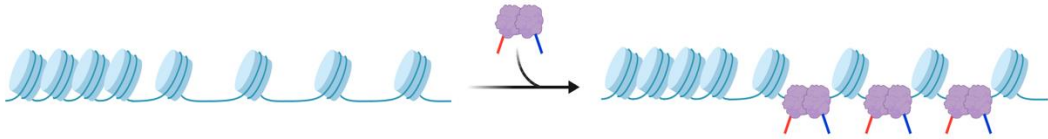
Have location need cell type
OR
Have cell type need better
location



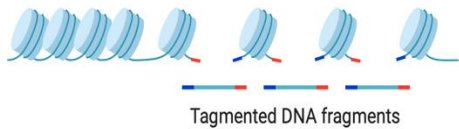




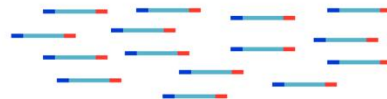
① Binding of Tn5 transposases to open chromatin



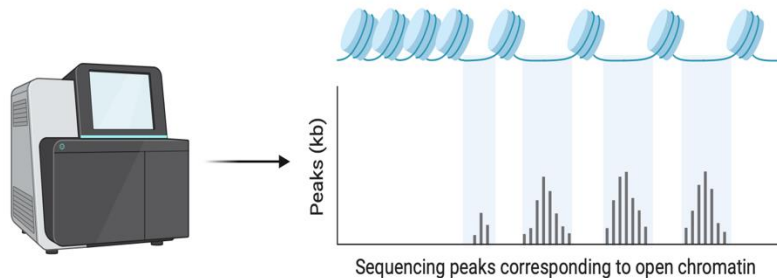
② Fragmentation and tagging (tagmentation) by Tn5 transposase



③ DNA fragment purification and PCR amplification



④ Next generation sequencing of PCR product and data analysis



Epigenomics - ATACseq

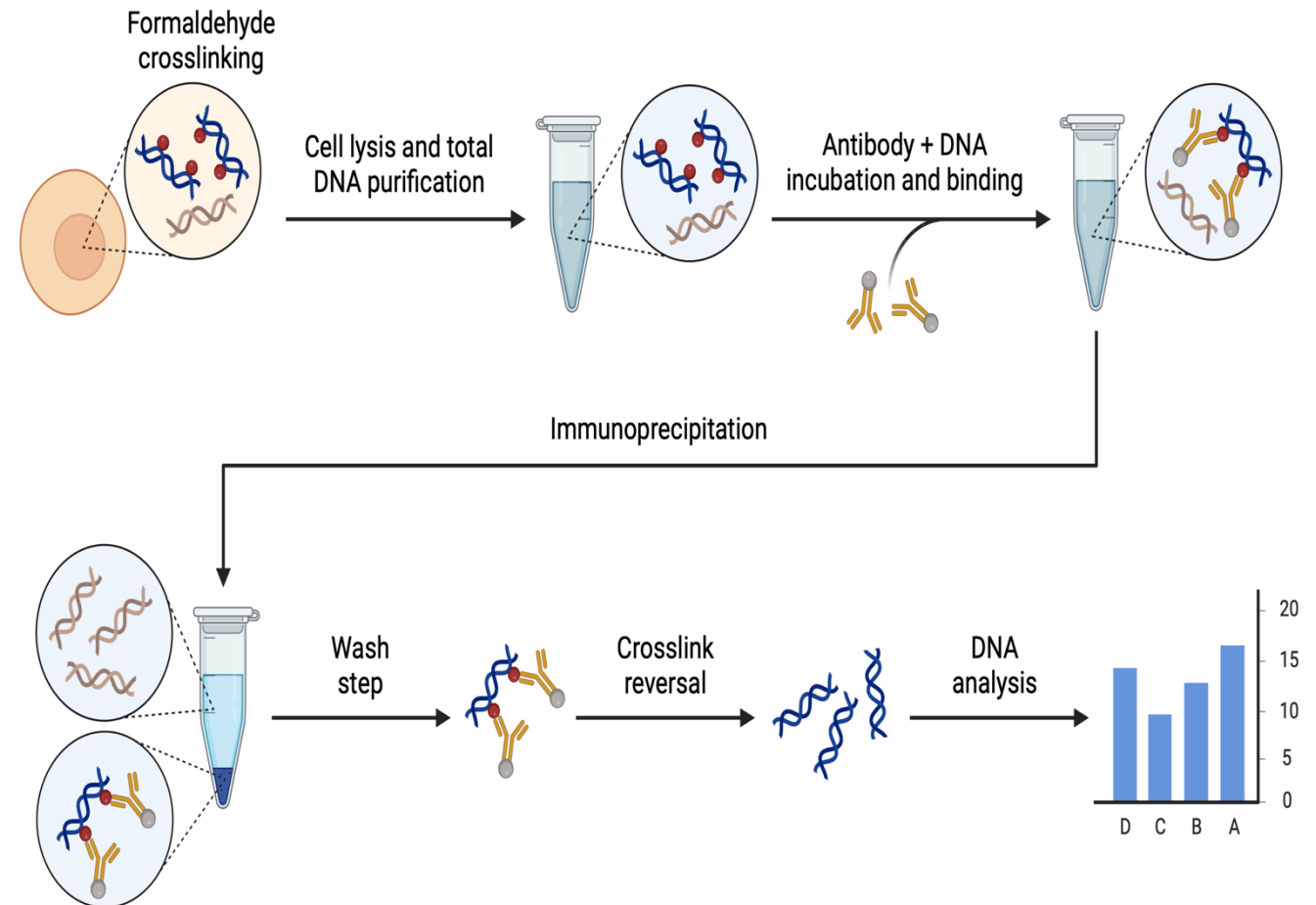
Assay of Transposase
Accessible Chromatin

Epigenomics

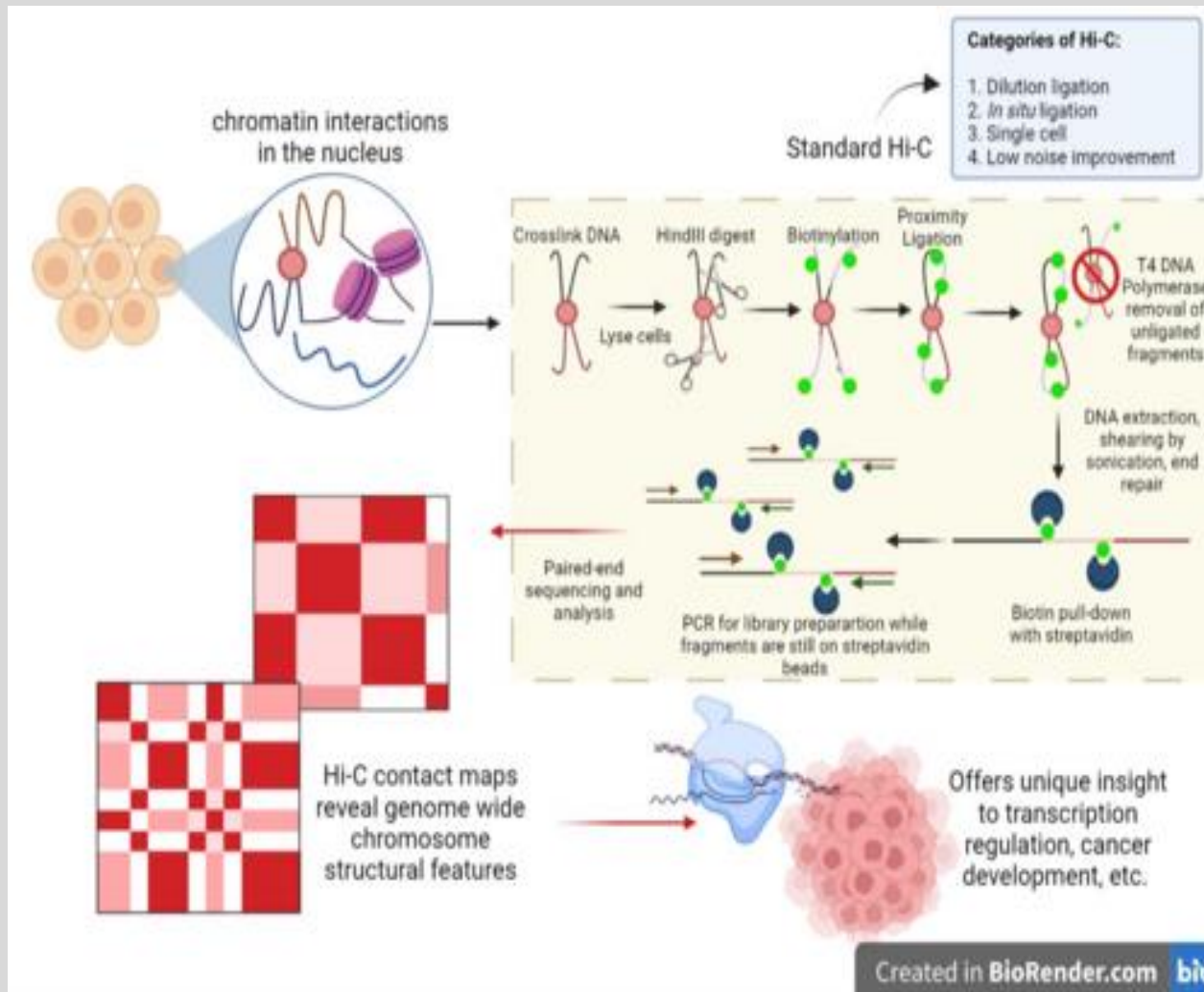
- ChIP-Seq

- Identifies the binding sites of DNA-associated proteins
- Maps global binding sites for a given protein (protein-DNA interactions)

Fast Chromatin Immunoprecipitation (ChIP) Protocol



Protein Purification by Hydrophobic Interaction Chromatography (HiC)

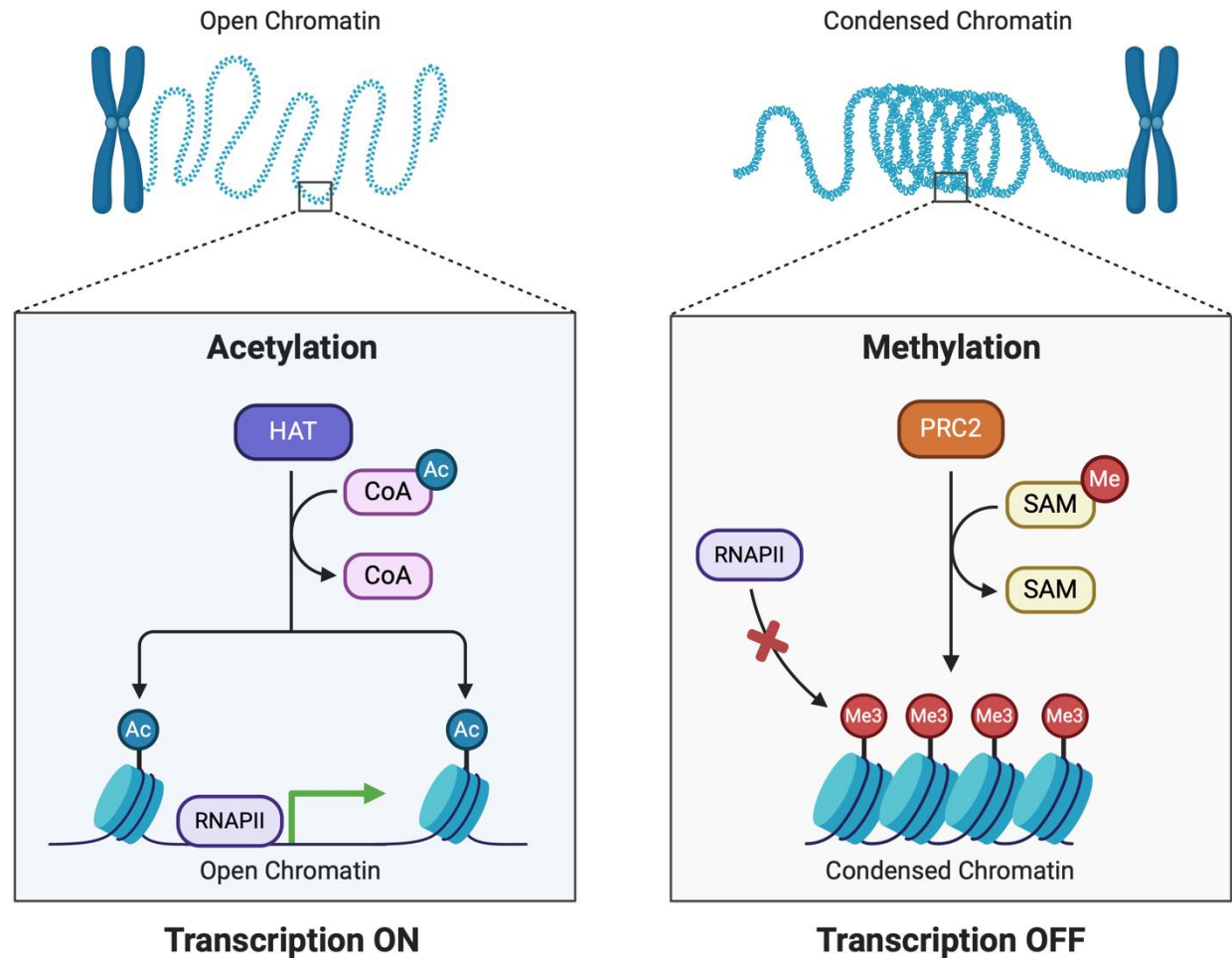


Epigenomics – Hi-C

- Long-range chromatin interactions

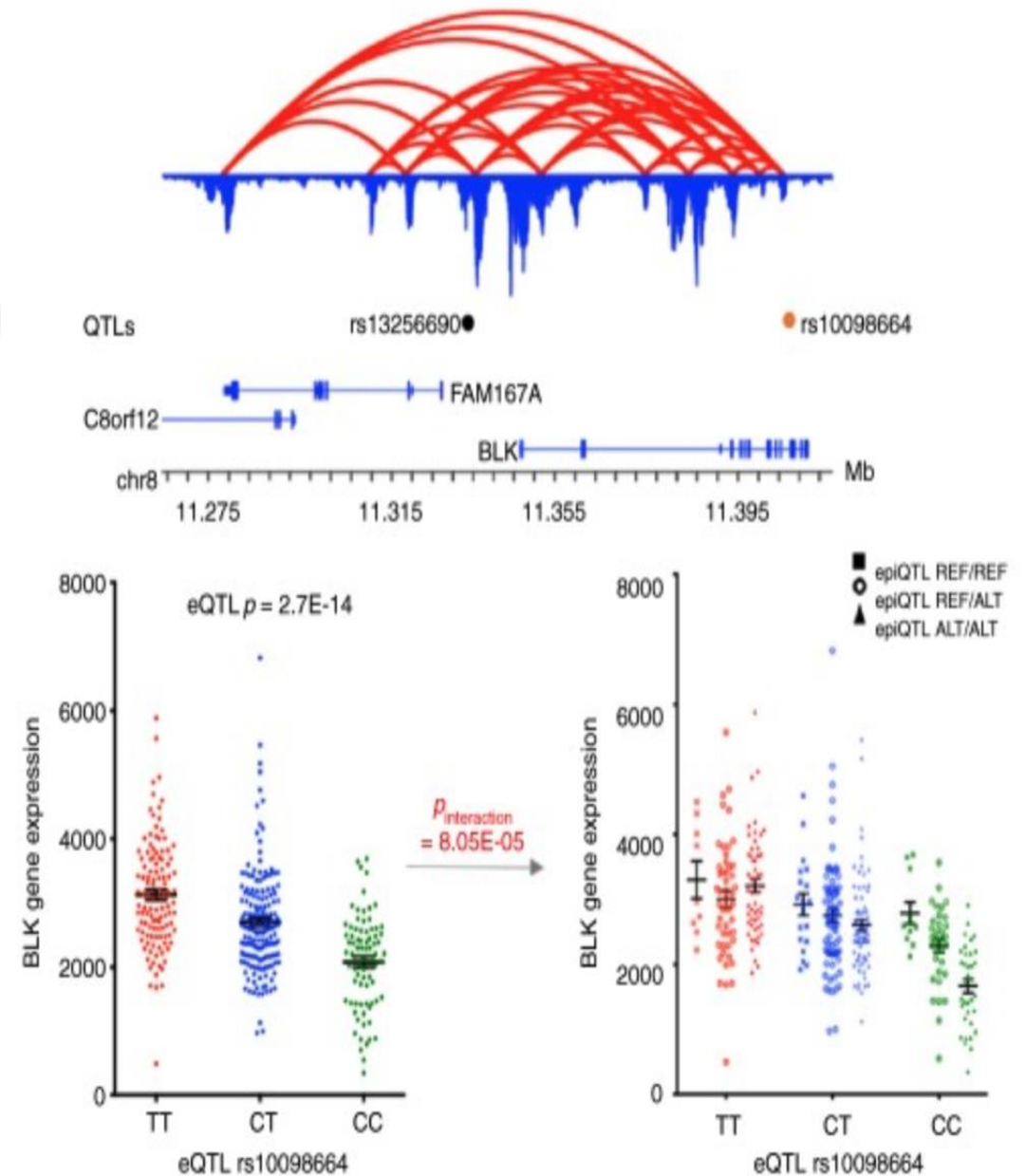
Epigenetics: Methylation

Affects gene expression
and therefore cell
differentiation and
behavior



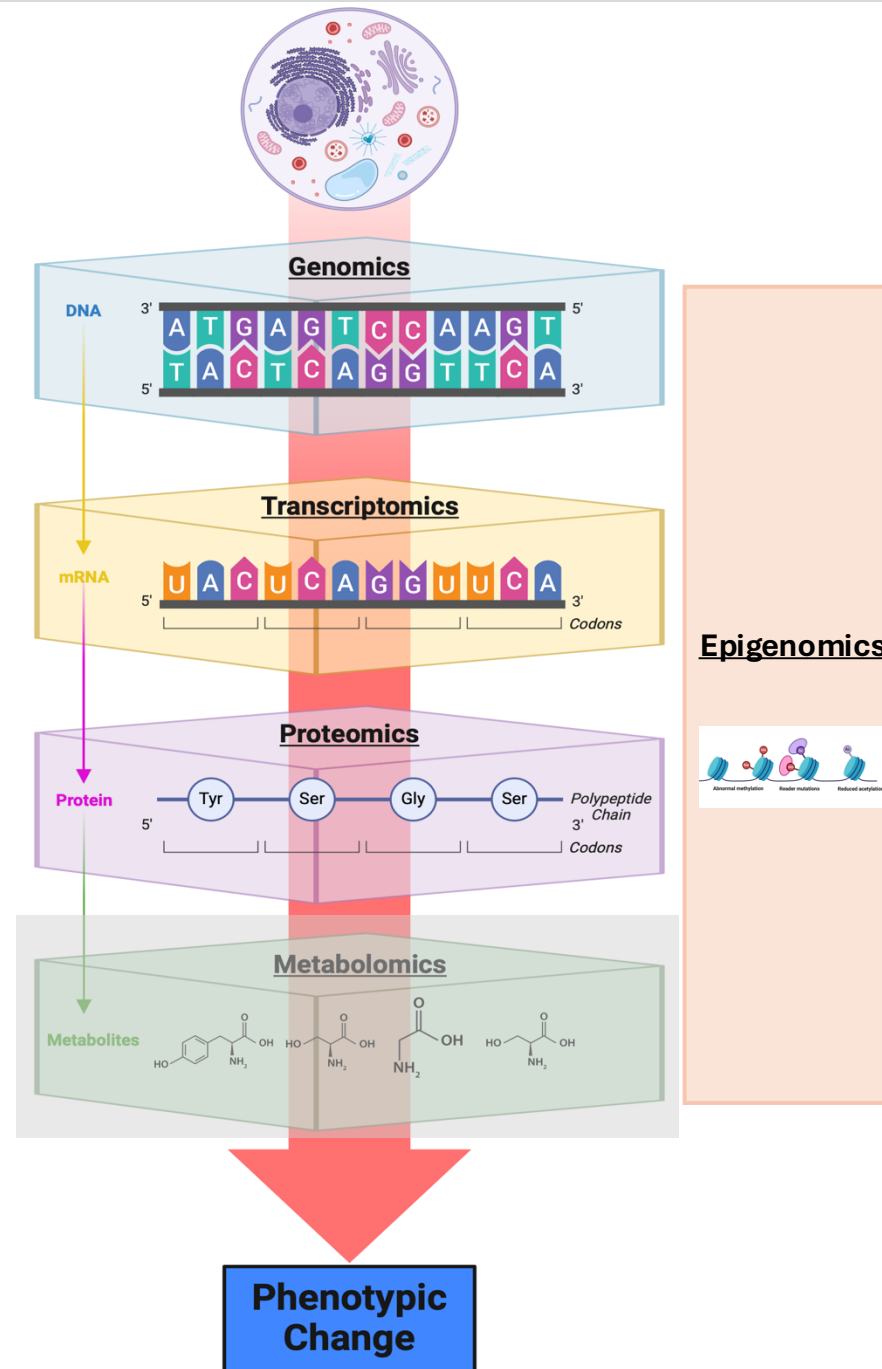
Able to Guide other Omic studies: Epigenetics

Example of *BLK* triplet interaction with eQTL rs10098664 and hQTLs rs13256690



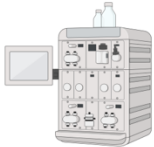
Pelikan RC, et al., Enhancer histone-QTLs are enriched on autoimmune risk haplotypes and influence gene expression within chromatin networks. *Nat Commun.*

2018 Jul 25;9(1):2905.



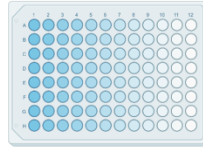
Proteomics Workflow and Techniques

1 Purification



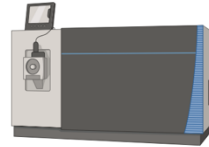
- Chromatography-based techniques

2 Detection



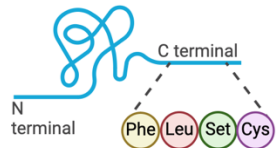
- ELISA
- Western blotting
- Protein microarray

3 Characterization



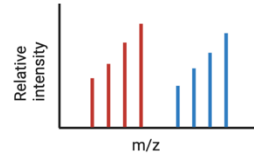
- Gel-based approaches
- Mass spectrometry

4 Sequence analysis



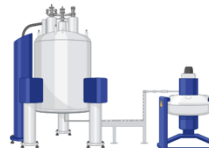
- Edman sequencing

5 Quantification



- ICAT
- SILAC
- iTRAQ

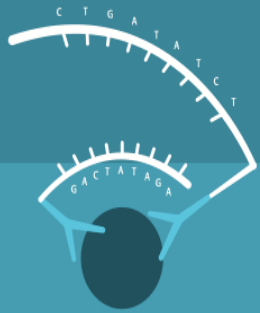
6 Structural analysis



- X-ray crystallography
- NMR spectroscopy

Proteomics

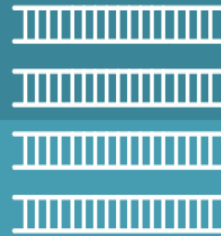
Proteomics – High Throughput



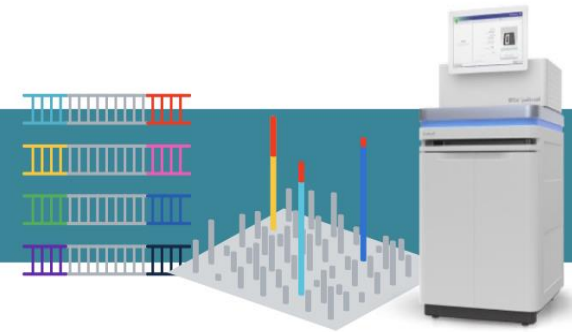
Antibody pairs, labelled with DNA oligonucleotides, bind target antigens in solution.



Oligonucleotides that are brought into proximity hybridize and are extended by a DNA polymerase.



This newly created double stranded DNA barcode is amplified by PCR, ready for readout by NGS or qPCR.

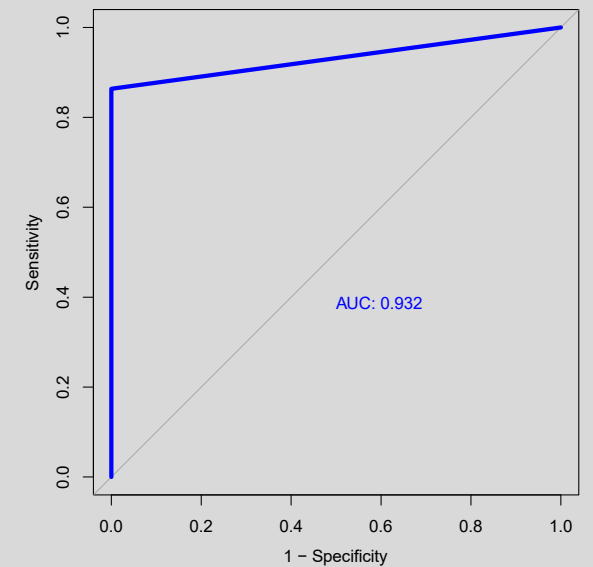
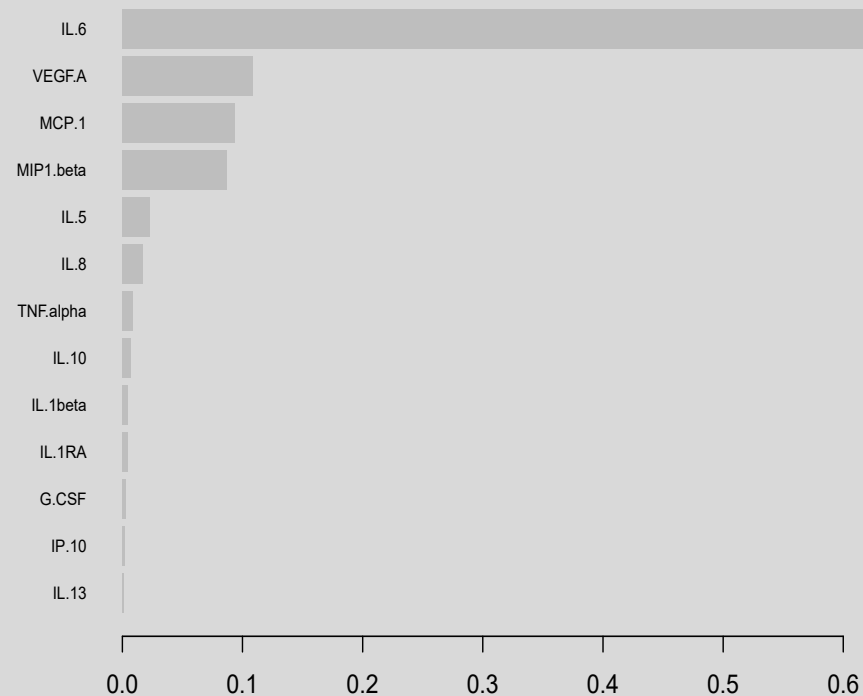
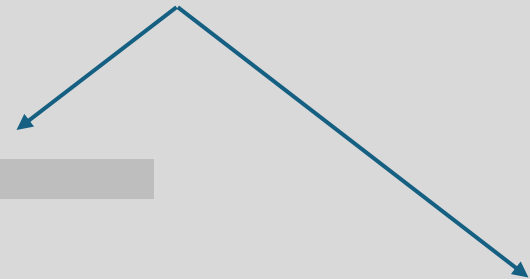
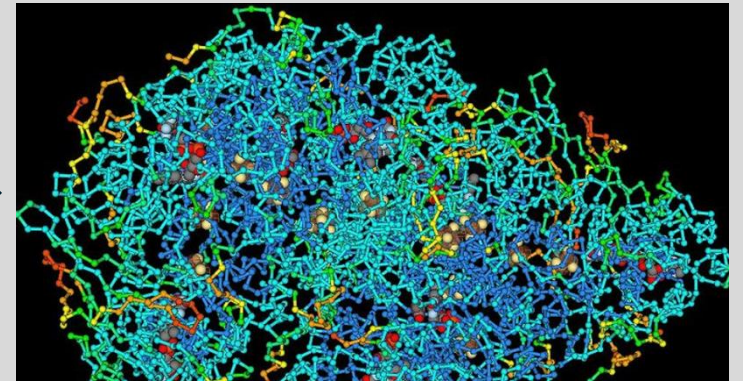


NGS readout for large-scale protein biomarker discovery

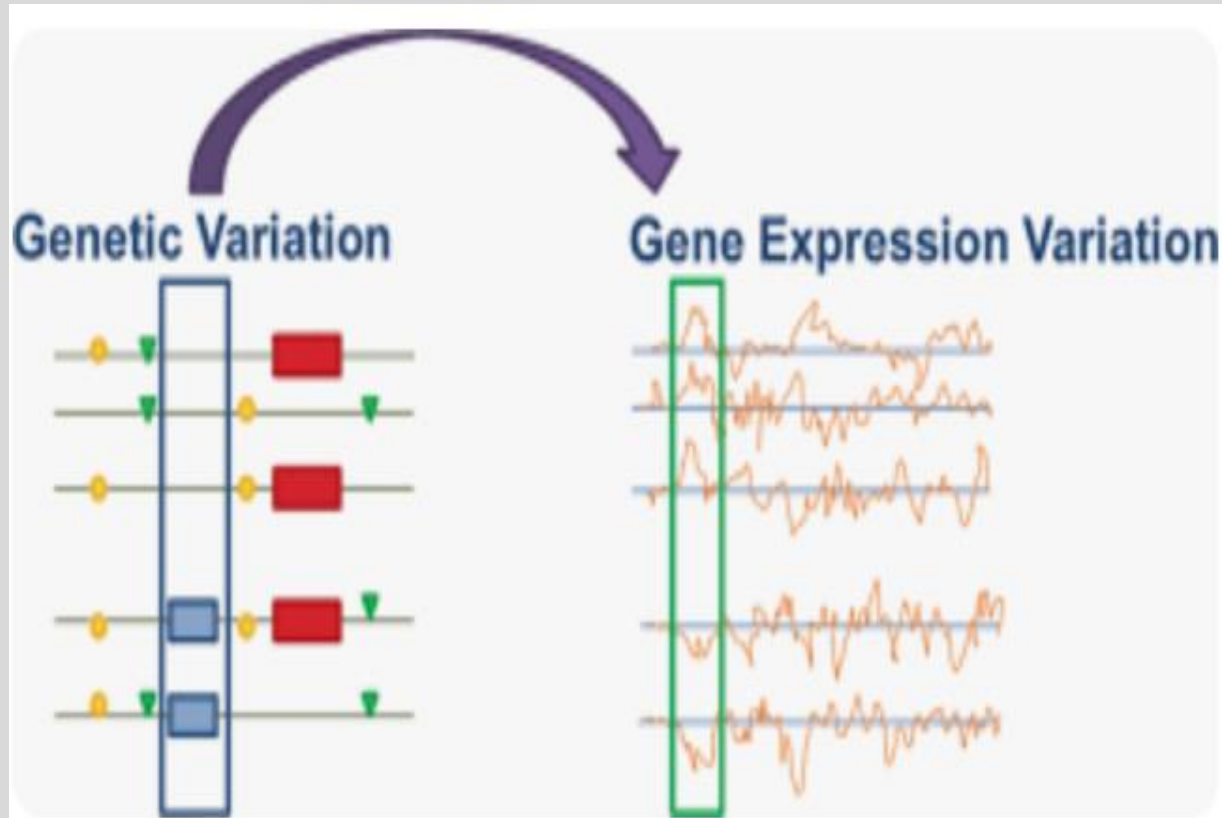


qPCR readout for targeted protein biomarker discovery and validation

Protein Modeling



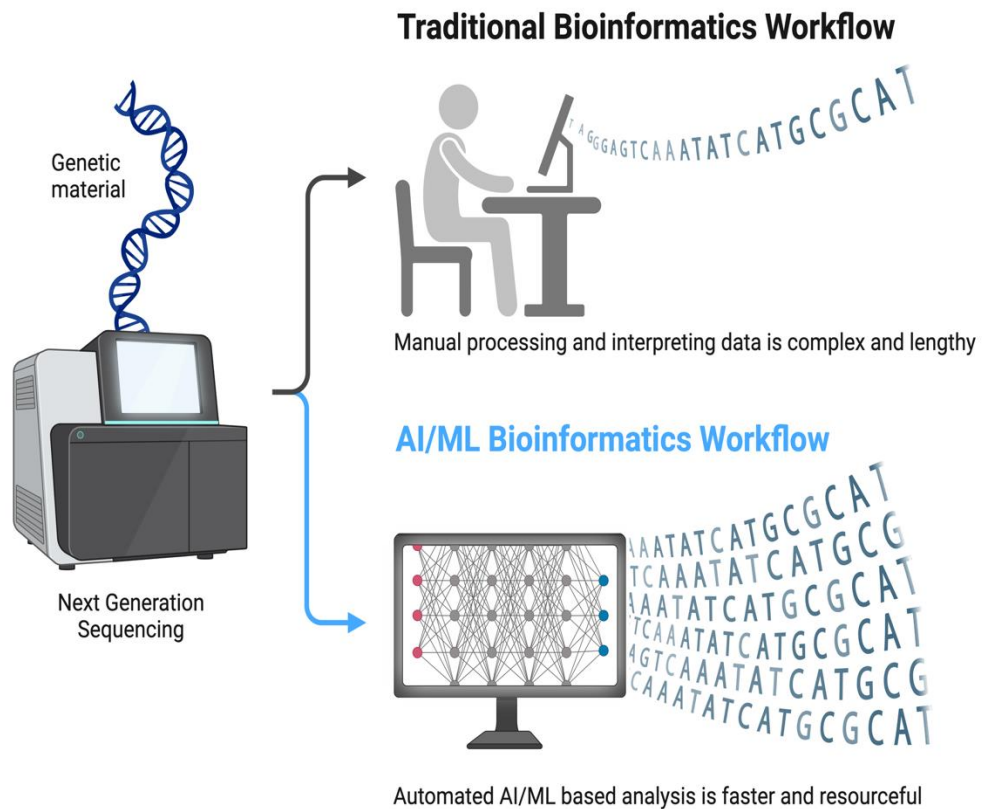
Multi-omic Integration



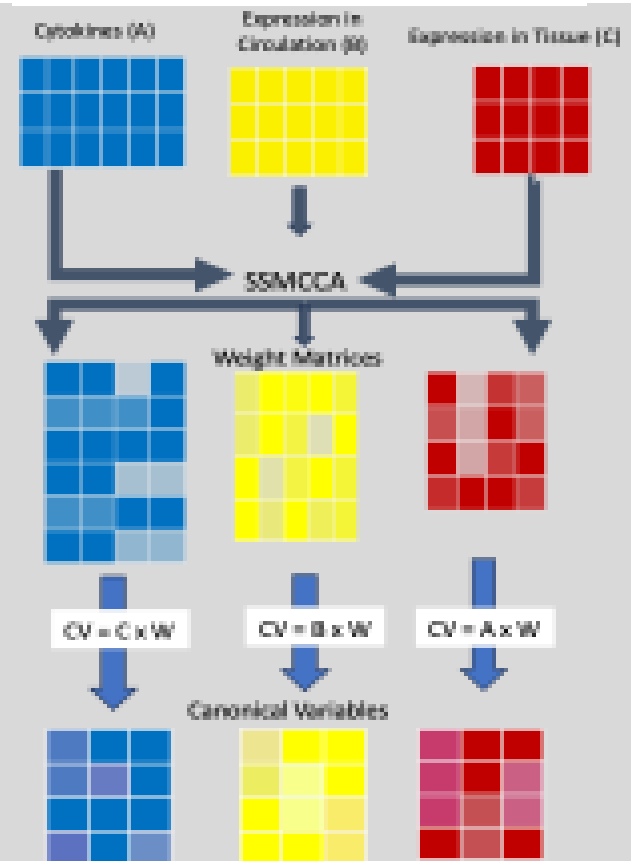
- Simplest examples, univariate:
 - eQTL
 - epiQTL
 - sQTL
 - pQTL

Nodzak, C. (2020). Introductory Methods for eQTL Analyses. In: Shi, X. (eds) eQTL Analysis. Methods in Molecular Biology, vol 2082. Humana, New York, NY.

Multi-omic Integration



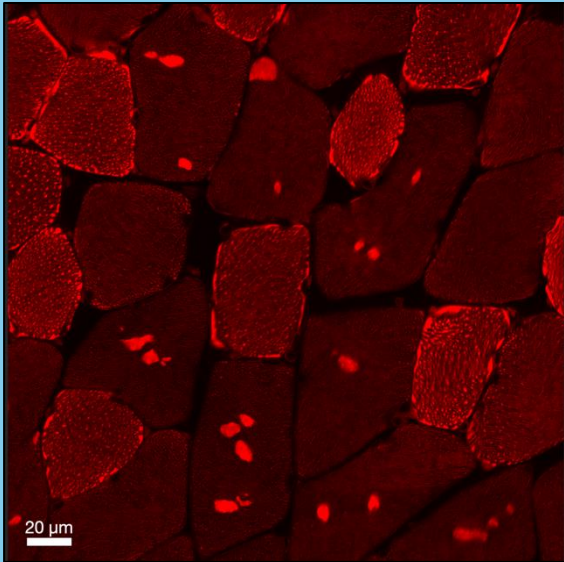
Sparse Supervised Multiple Canonical



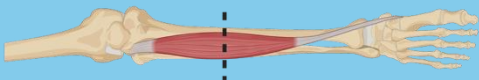
Jiang MZ, et al., Nhlbi TOPMed, TAWG. Canonical correlation analysis for multi-omics: Application to cross-cohort analysis. PLoS genetics. 2023;19(5):e1010517.

Advanced Microscopy

Skeletal Muscle Cell
Mitochondria (Red)

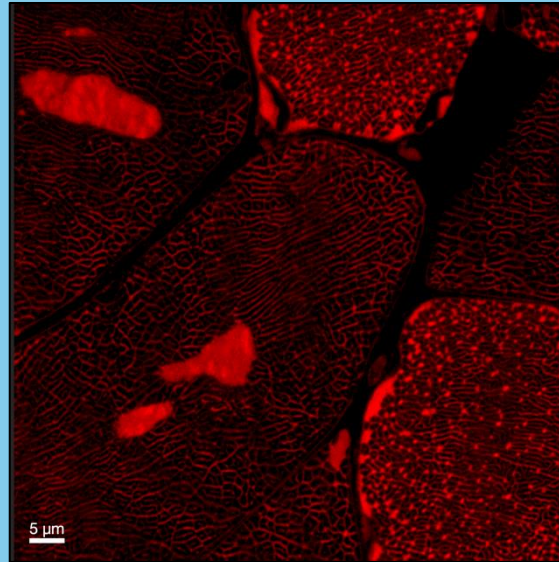


Front Lower Leg Muscle

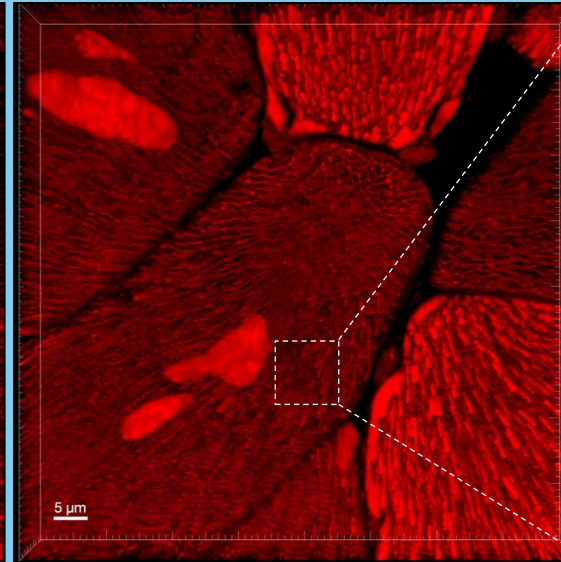


Cross-Sectional View
(~25 Muscle Fibers)

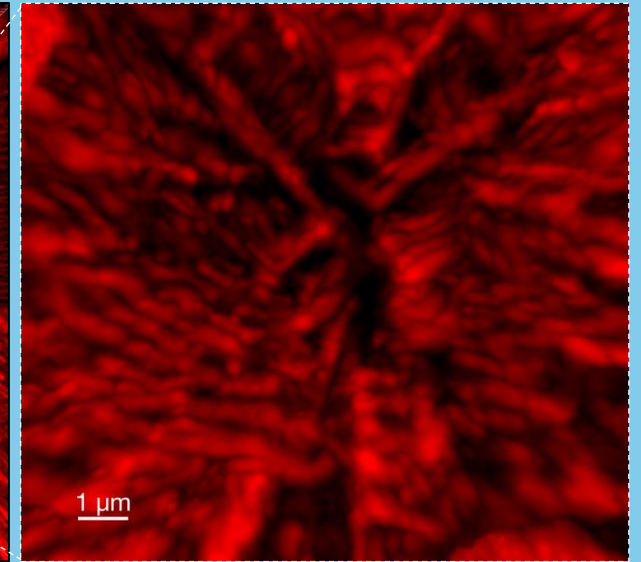
Super-Resolution (2D)



Super-Resolution (3D)



Skeletal Muscle Cell
Mitochondrial Reticulum



Intra- and Inter-Muscle
Cellular Mitochondria
Morphological
Differences (5-6 Fibers)

6-month-old
male mouse



32-year-old
male human

In-silico experimentation: Alpha-fold

- Automation of protein – protein interaction prediction

